



Biochip Course

Biosensors: Molecules

Marco Carminati

a.y. 2022/2023

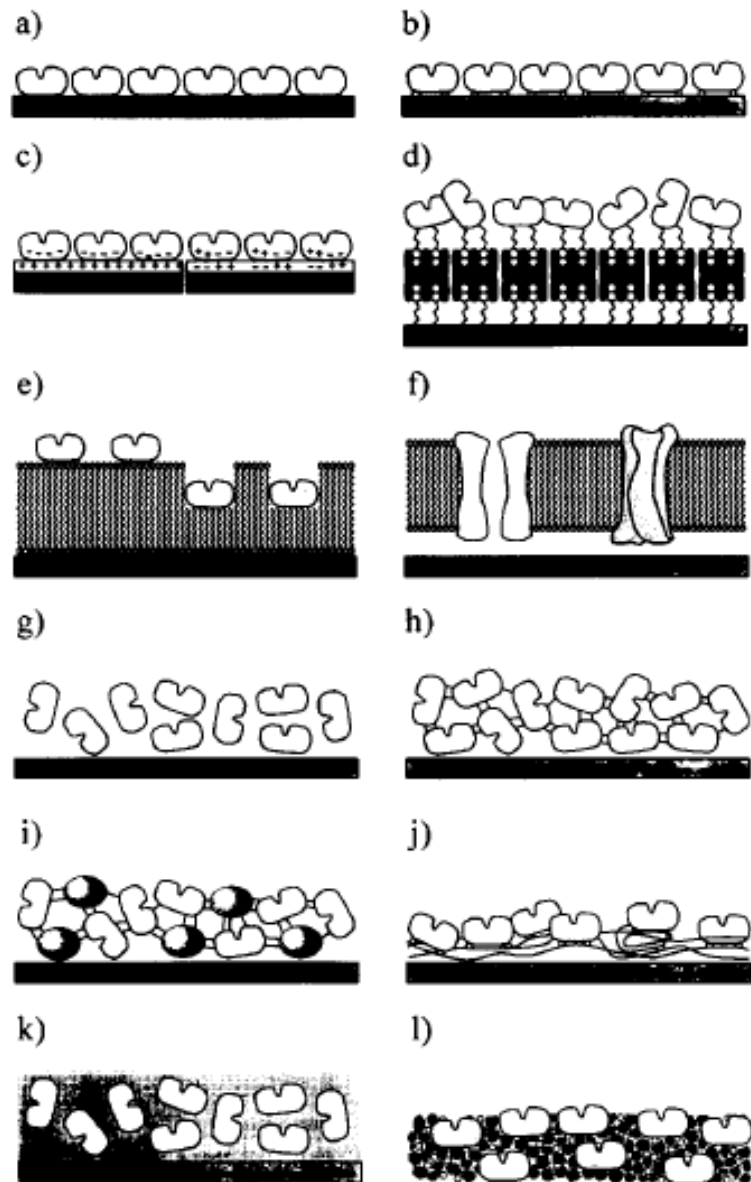
April 12th 2023



1. (How to extract the molecules: *sample preparation*)
2. How to attach the receptor molecules to a solid substrate
3. How to detect the molecules, **examples:**
 1. Micromechanical resonance (antibodies)
 2. Interface capacitance (DNA)
 3. Direct current detection (neurotransmitters)
 4. Enzyme-mediated current detection (glucose)
 5. Nanopores (DNA sequencing)
 6. Nanowires



Receptors Immobilization Strategies

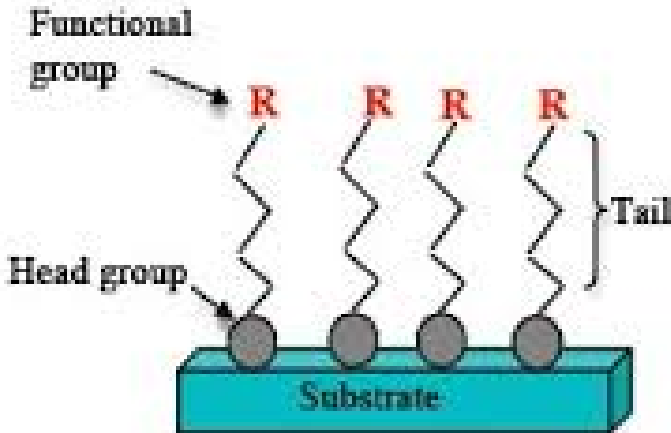


Molecular Affinity Biosensors

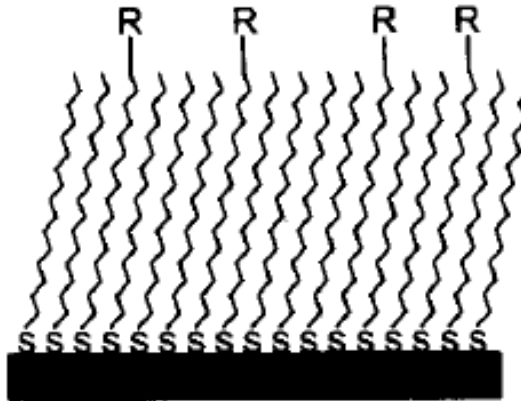
Fig. 2. Typical interfaces in biosensing: examples for the immobilisation of biological recognition units on transducer surfaces: (a) direct adsorption (physisorption), (b) covalent linking to the substrate, (c) adsorption by electrostatic forces, (d) coupling via biotin-avidin linkage, (e) adsorption at mono- or bilayers, (f) embedding in lipid bilayers, (g) entrapment behind a membrane, (h) cross-linking of biological molecules, (i) cross-linking by other large molecules (e.g. BSA), (j) covalent linking to polymer chains, (k) entrapment within a gel of a polymer (non-conducting medium), (l) entrapment by mixing with carbon paste, conducting polymers or organic salts (conducting medium).



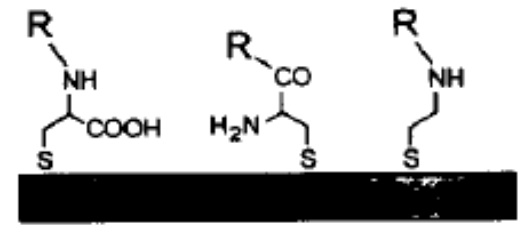
Linkers: Self-Assembled Monolayers



a) Alkanethiols



b) Cysteines, Cysteamines



Trade-off chain length:

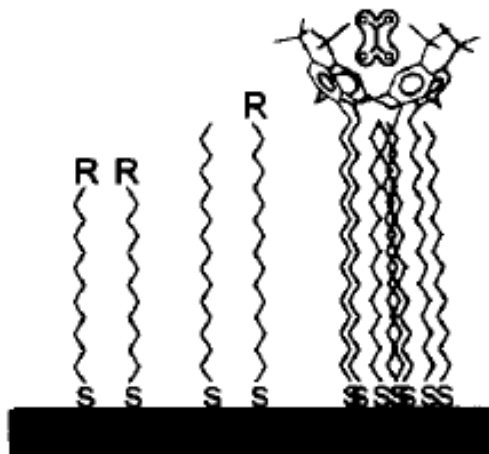
Short

- Good charge transfer
- Less risk of bending

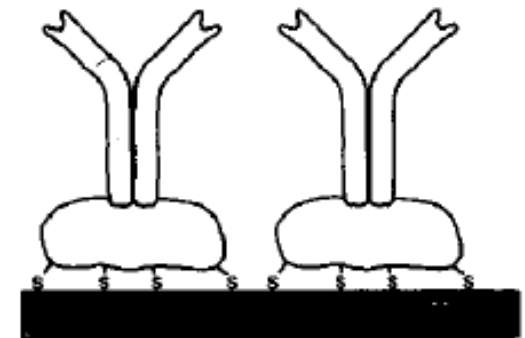
Long

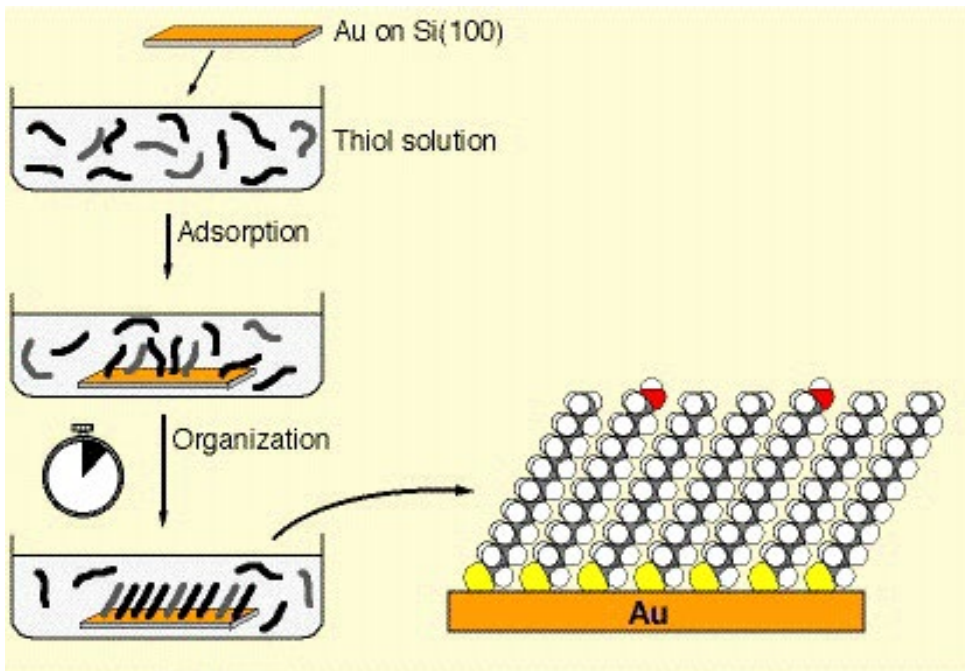
- Less interaction with the substrate

c) Disulphides



d) Protein A with antibodies





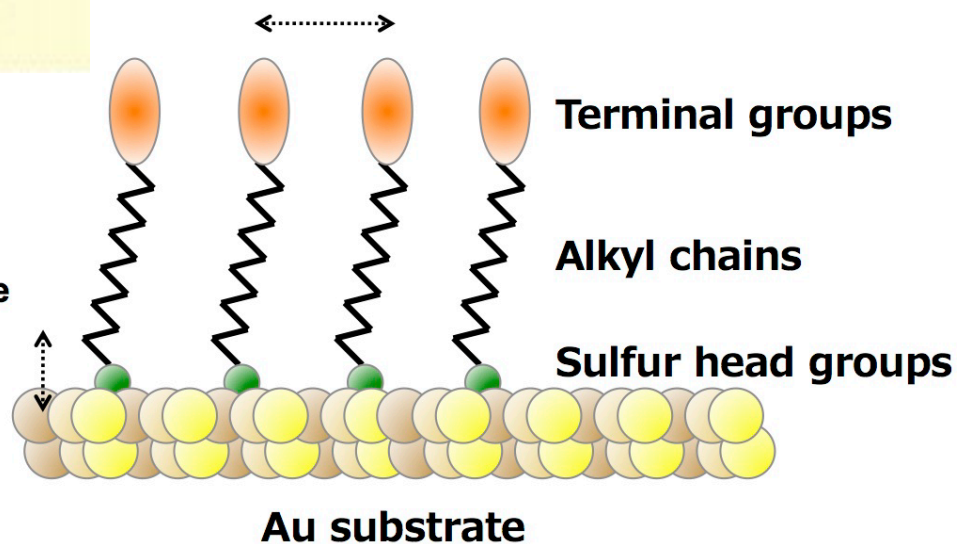
Sulfur head binds to gold

Intermolecular interactions

interactions between
terminal groups
alkyl chains
sulfur head groups

Substrate-molecule interaction

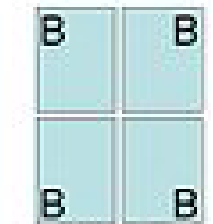
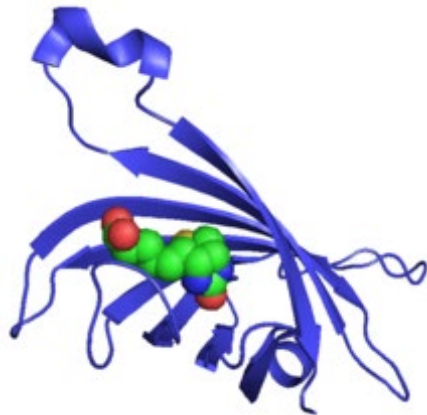
interactions between
sulfur and Au surface





Basic Bricks: Streptavidin - Biotin

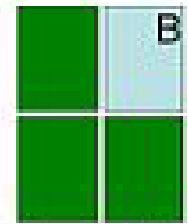
Streptavidin: protein extracted from a bacterium



Wild-type streptavidin



Monomeric streptavidin



Monovalent streptavidin
(Dead subunits in green)

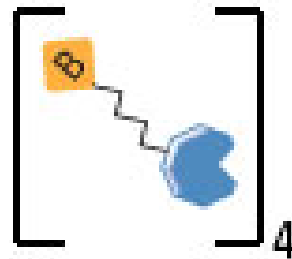
Most resistant and robust (non-covalent) binding in nature

Biotin – Streptavidin molecules



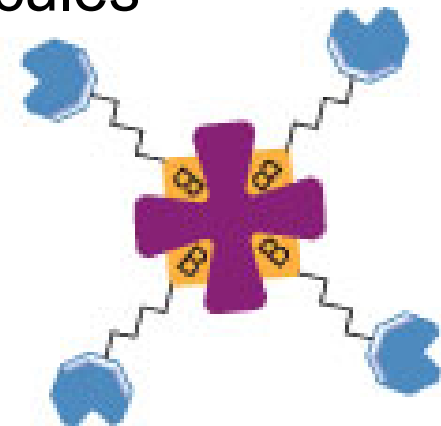
Avidin

+



Biotin conjugate

=



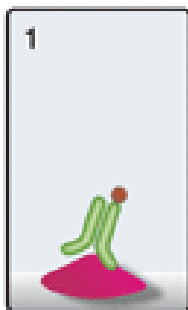
Avidin-Biotin Complex



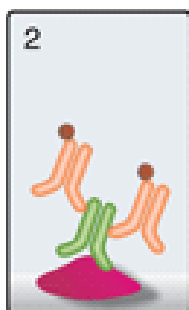
Affinity used for sandwiches of molecules



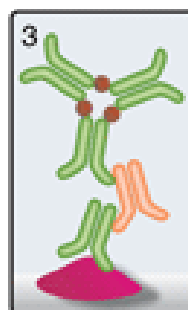
Labeled Primary Antibody Method



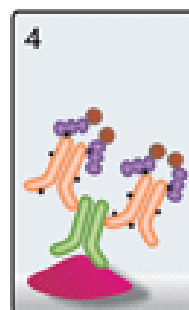
Labeled Secondary Antibody Method



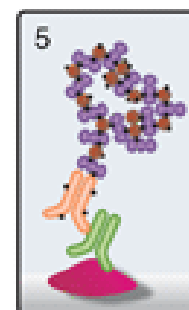
Peroxidase Antiperoxidase (PAP) Method



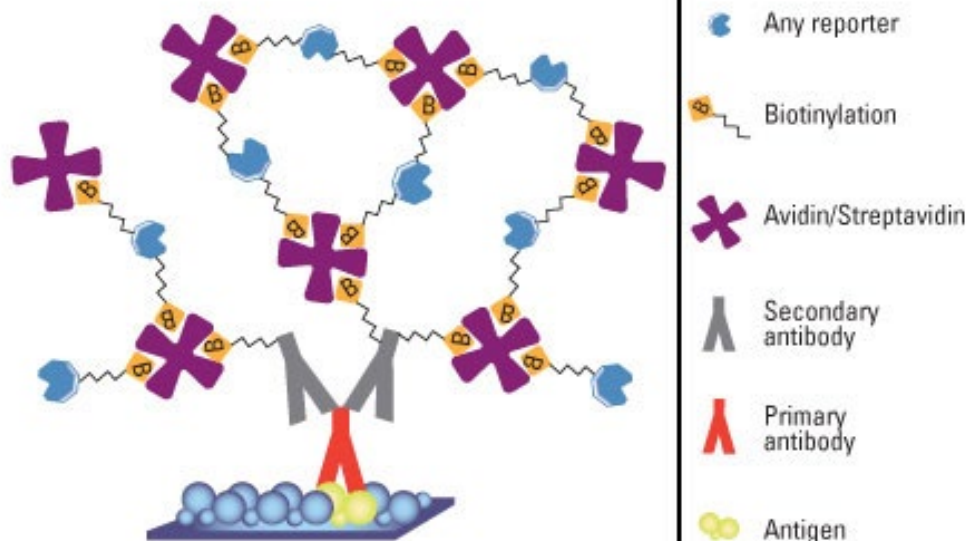
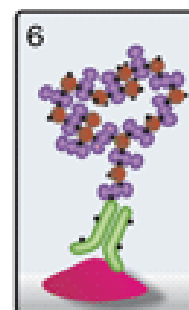
Labeled Avidin Method



Avidin/Biotinylated Enzyme Complex (ABC) Method



Avidin/Biotinylated Enzyme Complex (ABC) Method





Summary of Detection Techniques

Mechanical Resonance

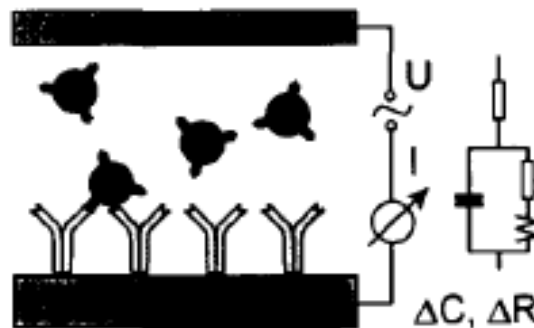
a)



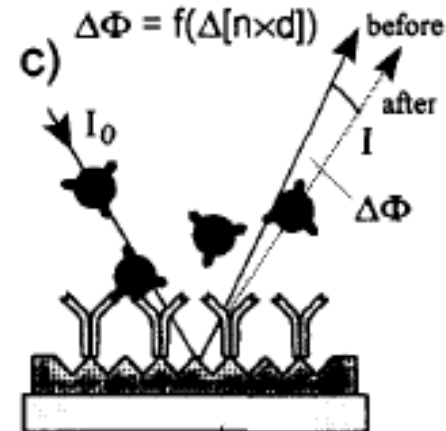
$$\Delta f \sim \Delta m$$

Electrochemical

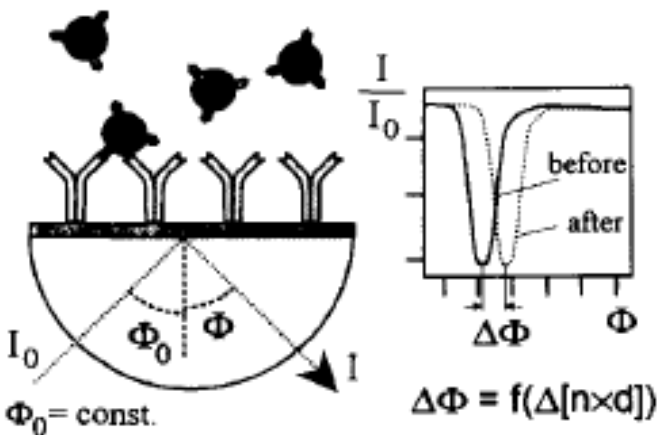
b)



$$\Delta C, \Delta R$$

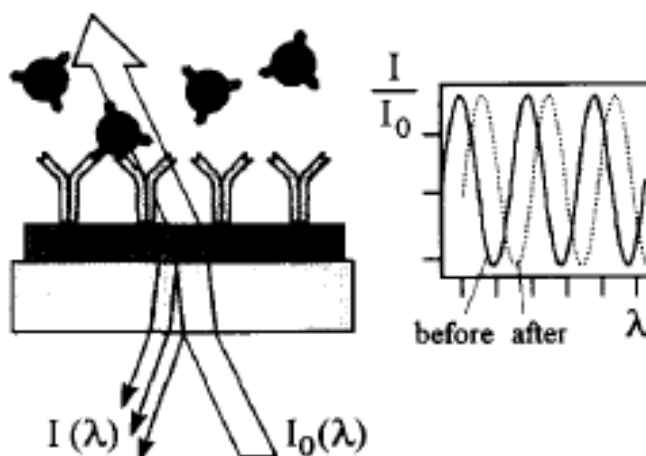


d)

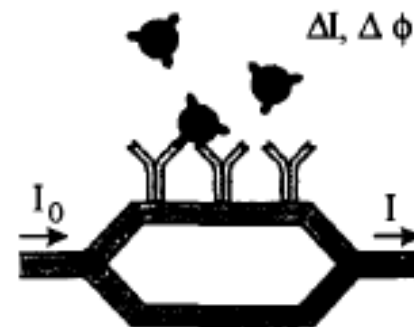


SPR

e)



f)

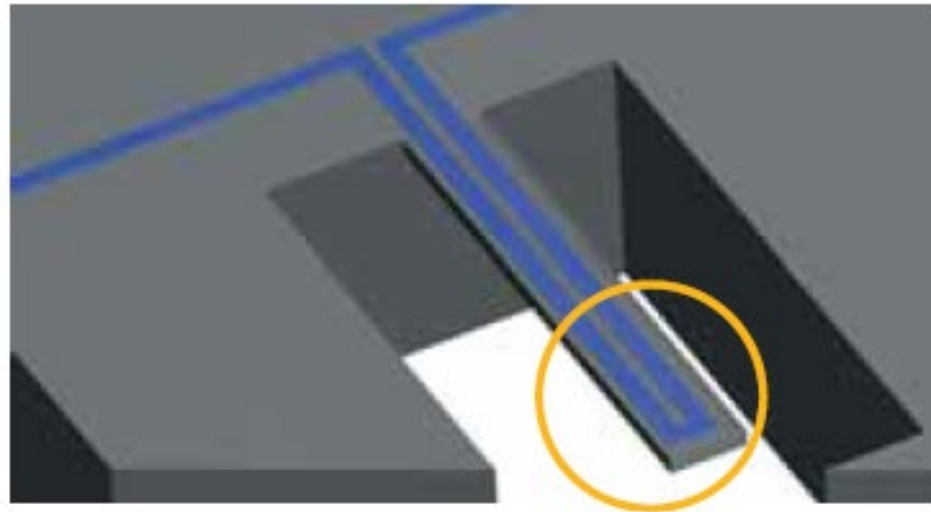


Evanescent



1. Mechanical Detection: Weighing Molecules

Leveraging mass-sensitive micromechanical resonance:

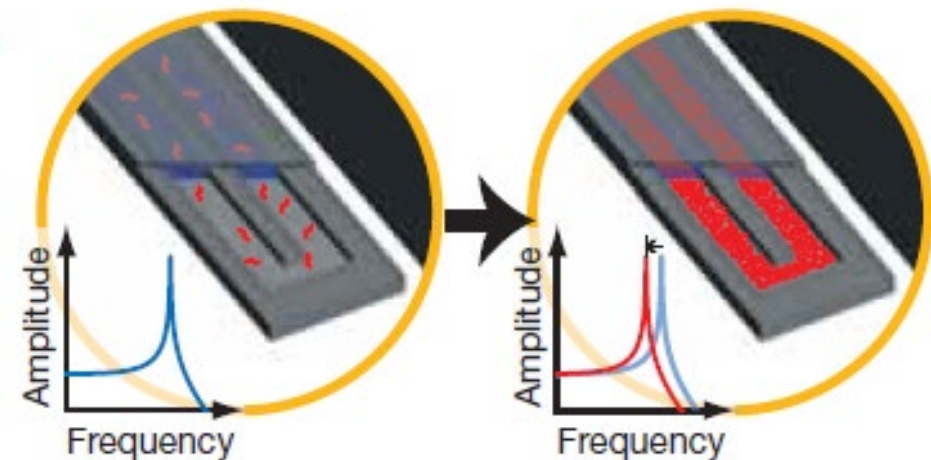


Shift in the resonance frequency

spring constant

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m^* + \alpha \Delta m}}$$

mass position extra
coeff. mass



nature

Vol 446 | 26 April 2007 | doi:10.1038/nature05741

LETTERS

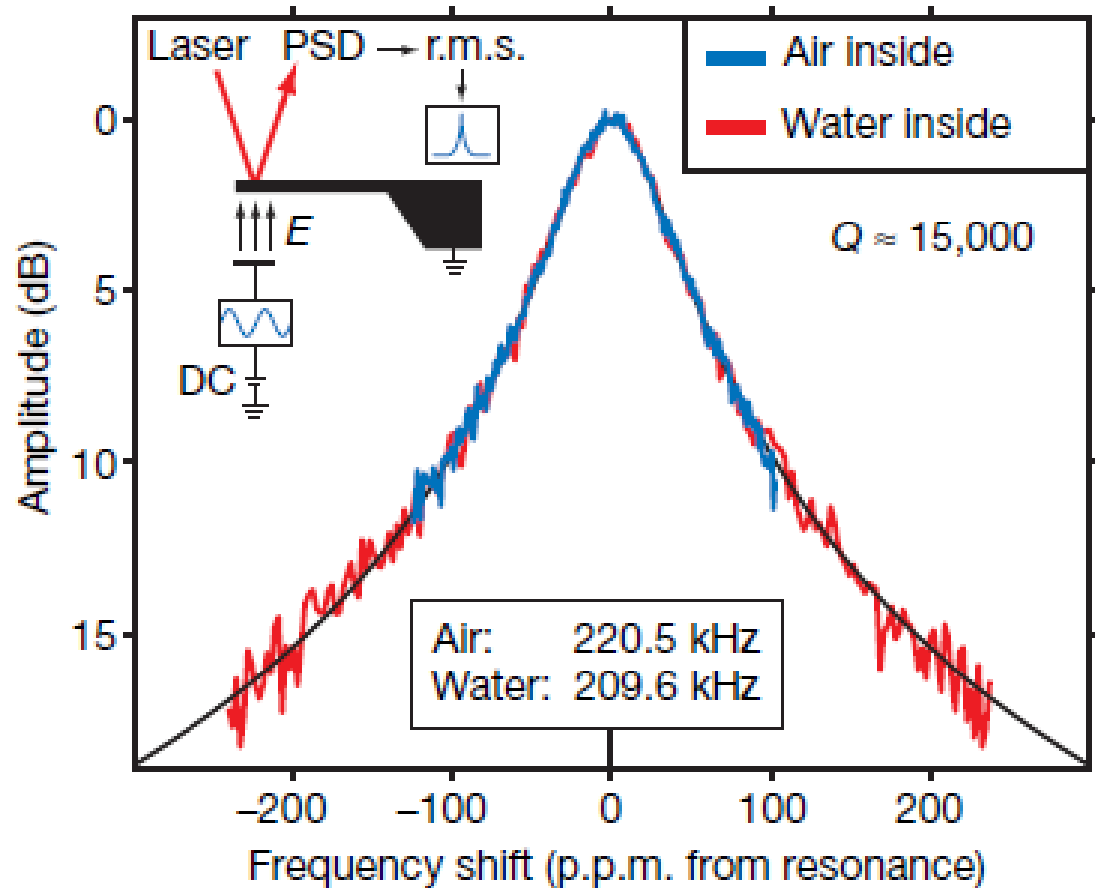
Weighing of biomolecules, single cells and single nanoparticles in fluid

Thomas P. Burg^{1*}, Michel Godin^{1*}, Scott M. Knudsen¹, Wenjiang Shen³, Greg Carlson³, John S. Foster³, Ken Babcock^{3,4} & Scott R. Manalis^{1,2}



Realization

Suspended Silicon cantilever: $200 \times 33 \times 7 \mu\text{m}$



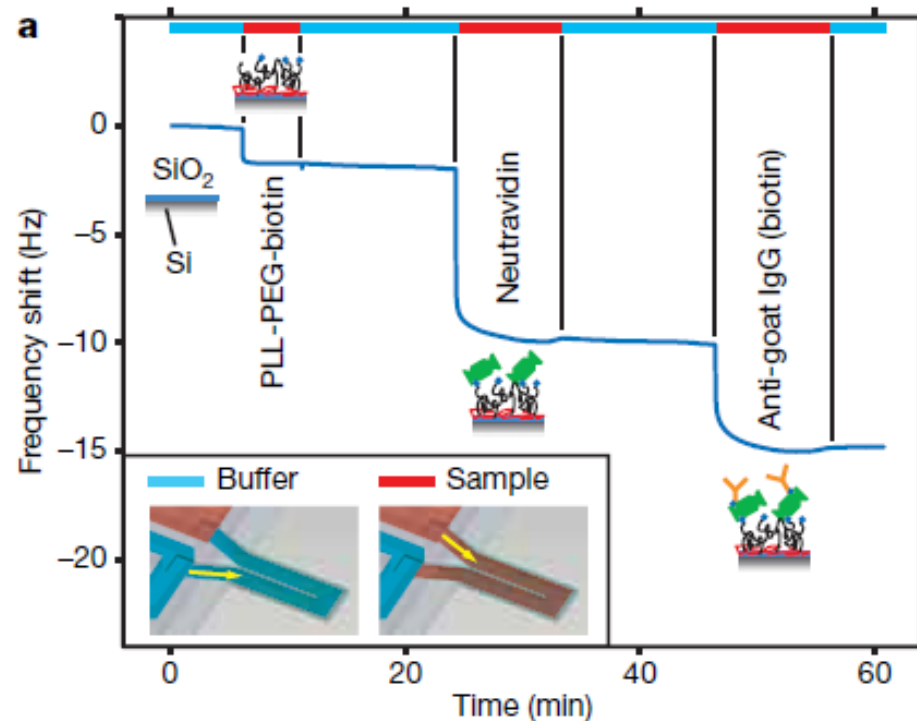
Oscillator stability 10ppb!



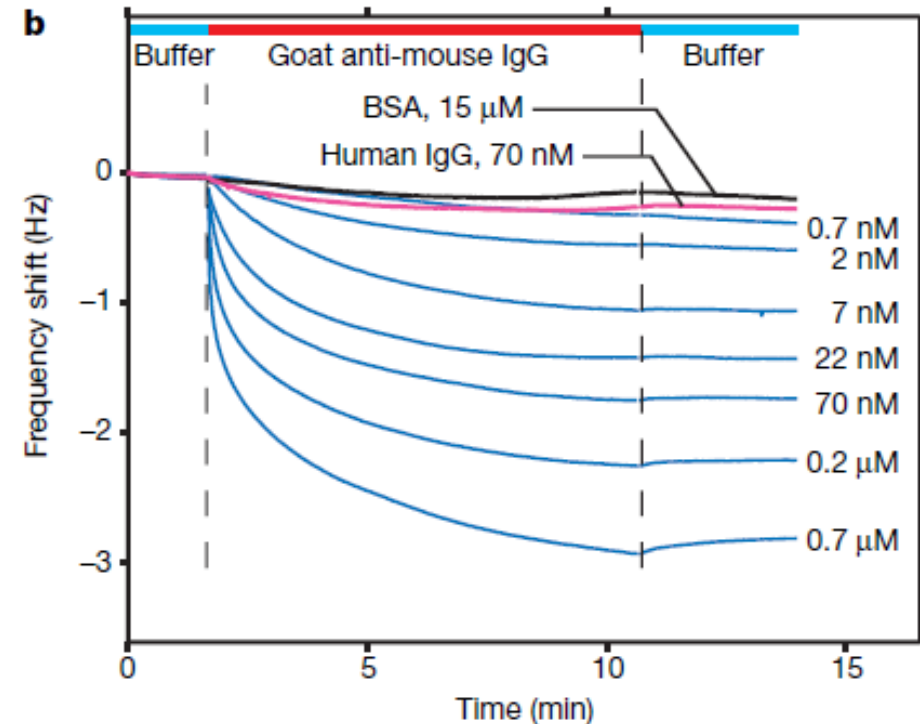
Example of Detection

Not selective: the specificity lies in the receptor (antibody)

Receptor deposition

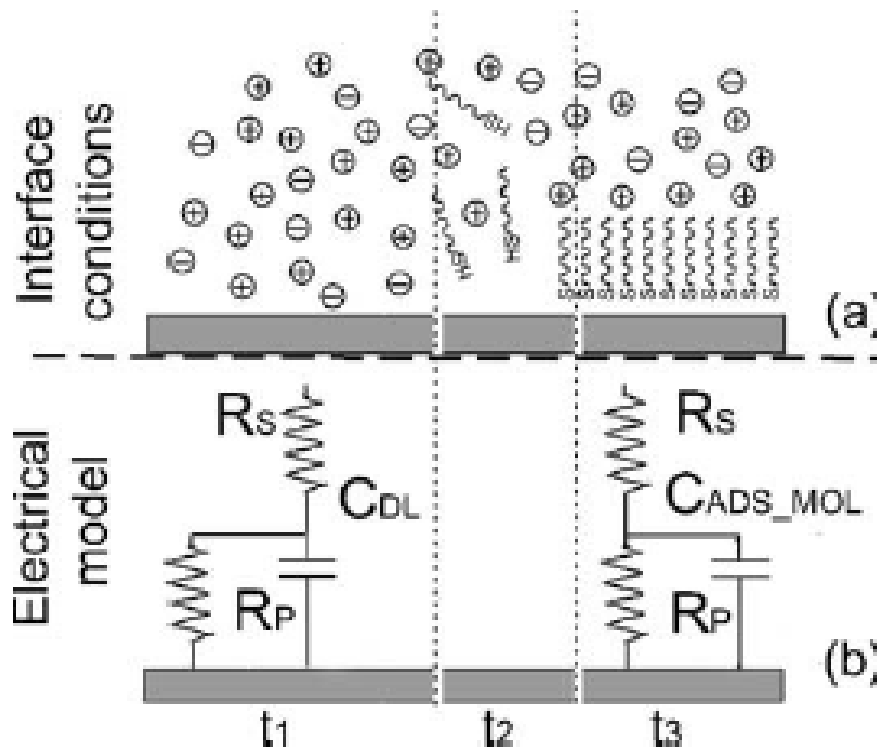


Detection



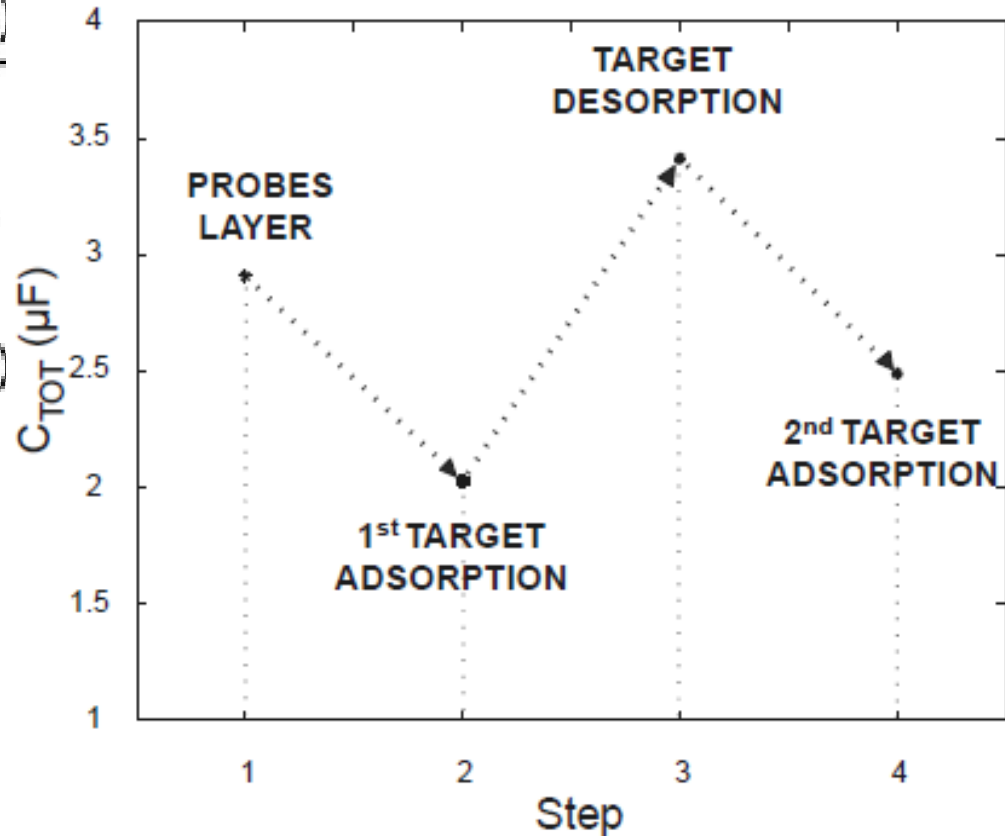
Sensitivity: 0.01ng/cm² – 1fg total (SPR 0.05ng/cm² – 1pg)

2. Capacitive Detection: DNA



DNA detection by integrable electronics

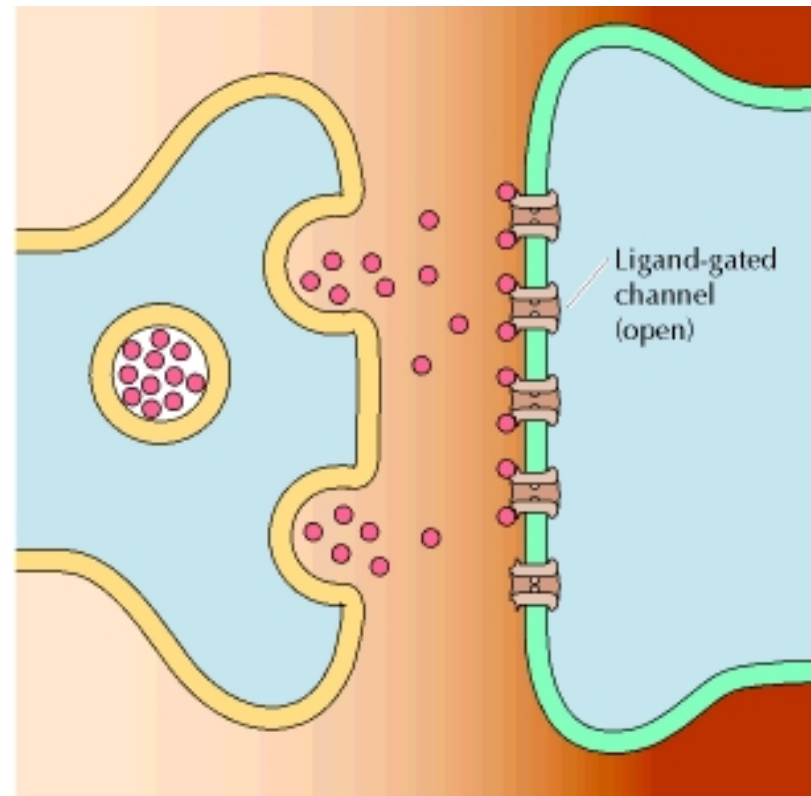
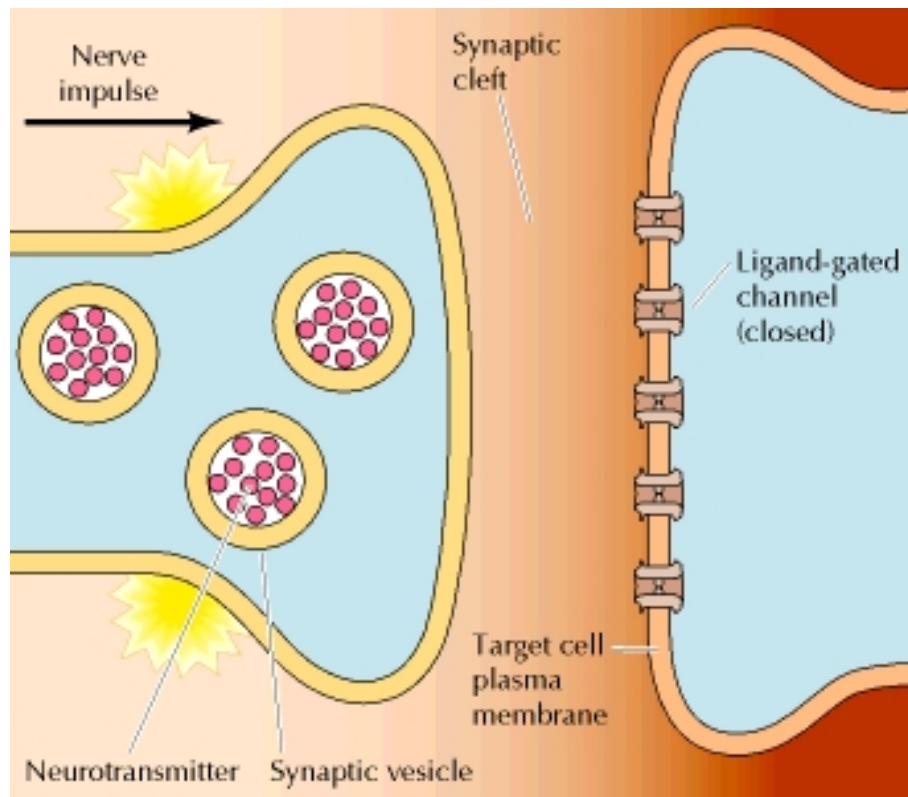
C. Guiducci^{a,*}, C. Stagni^a, G. Zuccheri^b, A. Bogliolo^c,
L. Benini^a, B. Samorì^b, B. Riccò^a





The Synapse

Chemical mediation: $5 \cdot 10^{14}$ synapses in the human brain

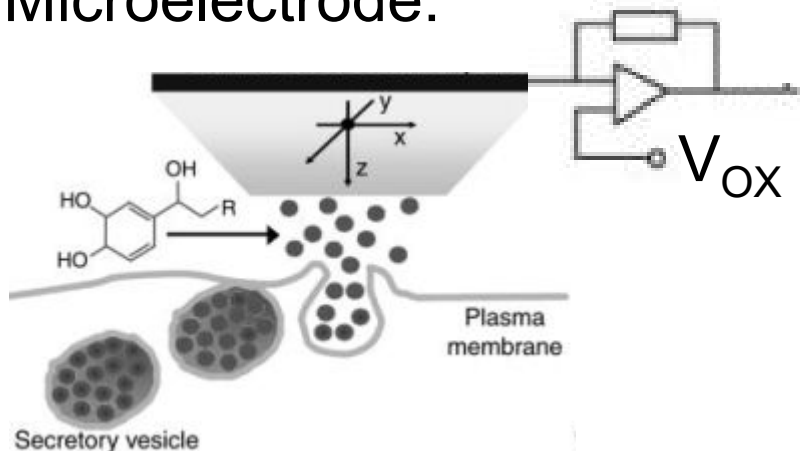


Exocytosis: extra-cellular release of small molecules in vesicles



3. Amperometric Detection of Molecules

Microelectrode:

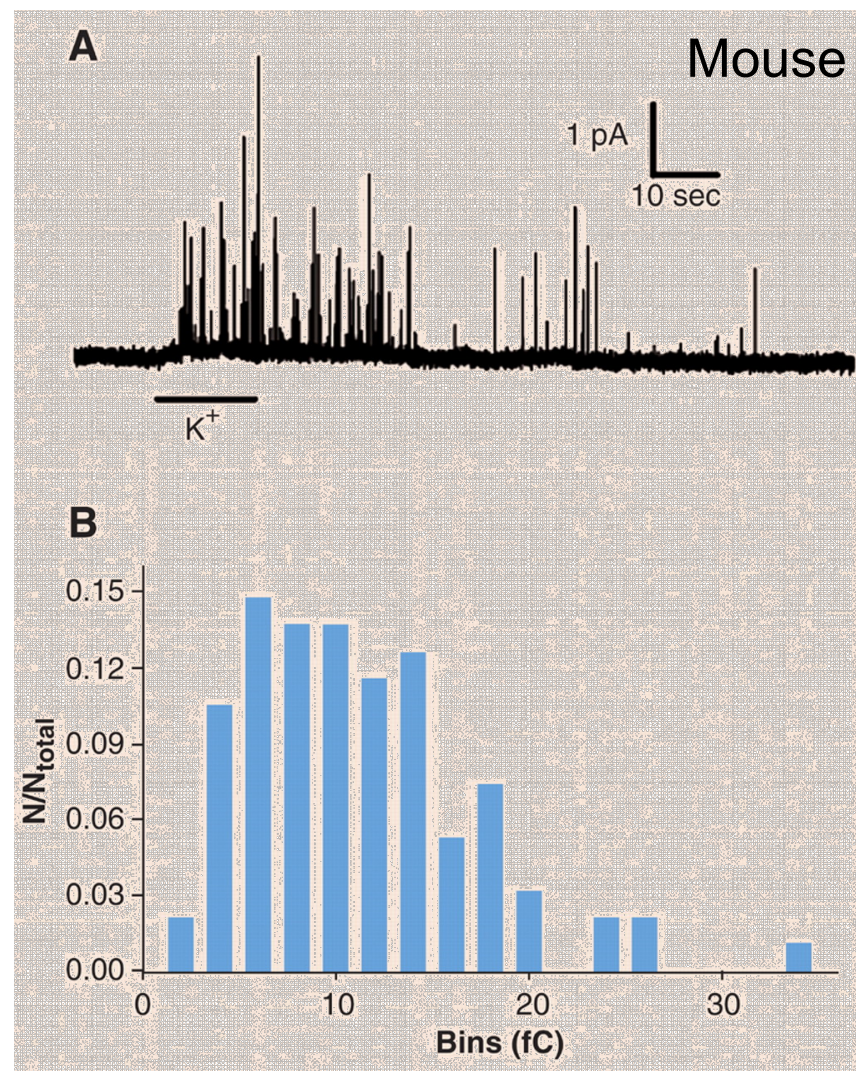


Chemical messengers:

- Dopamine, adrenaline etc..

Current tracking:

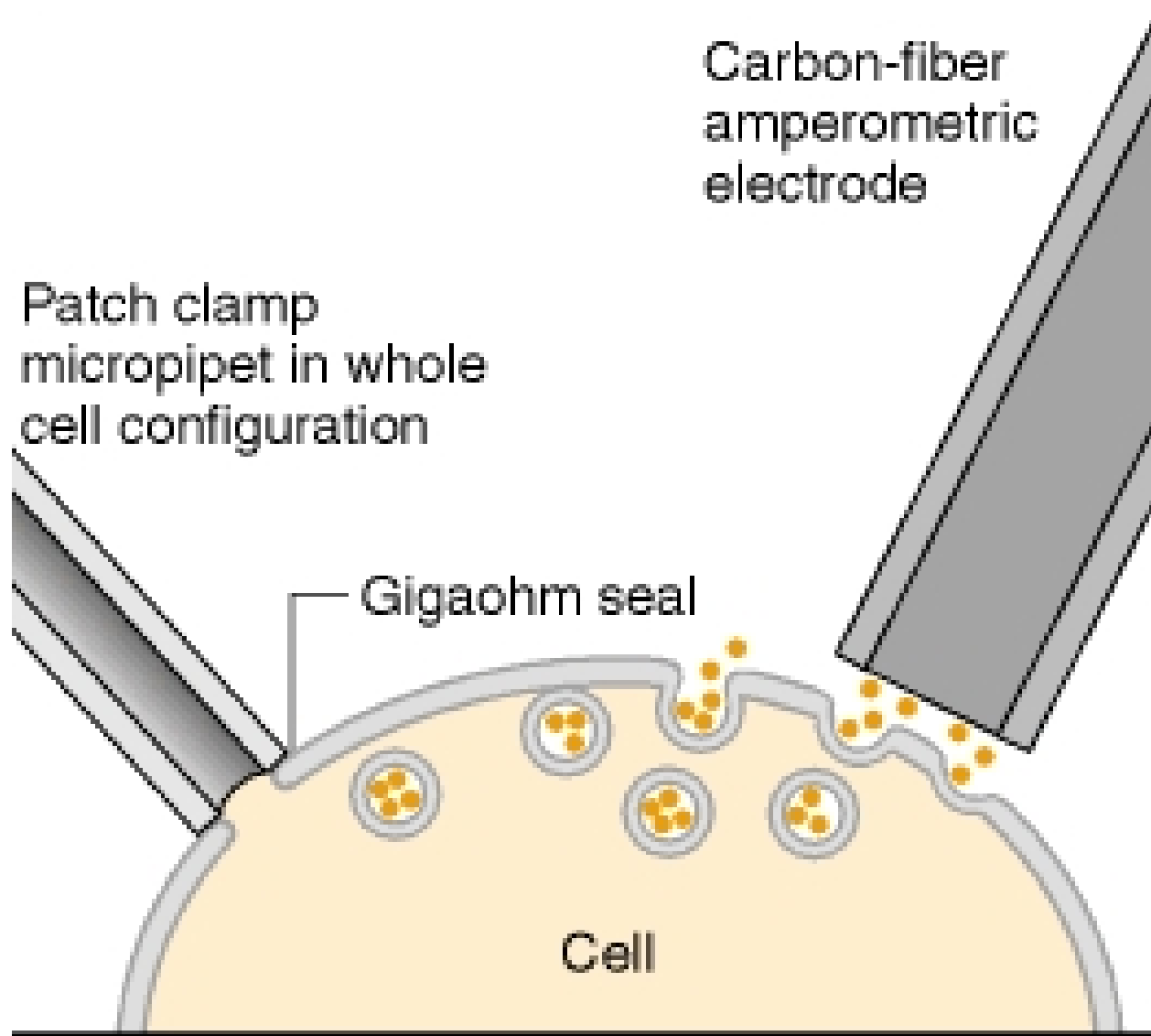
- $5 \cdot 10^4$ molecules in a vesicle
- 10fC released in 1ms $\rightarrow$ **~ 10 pA**



M. Wightman, *Science* **17** 2006



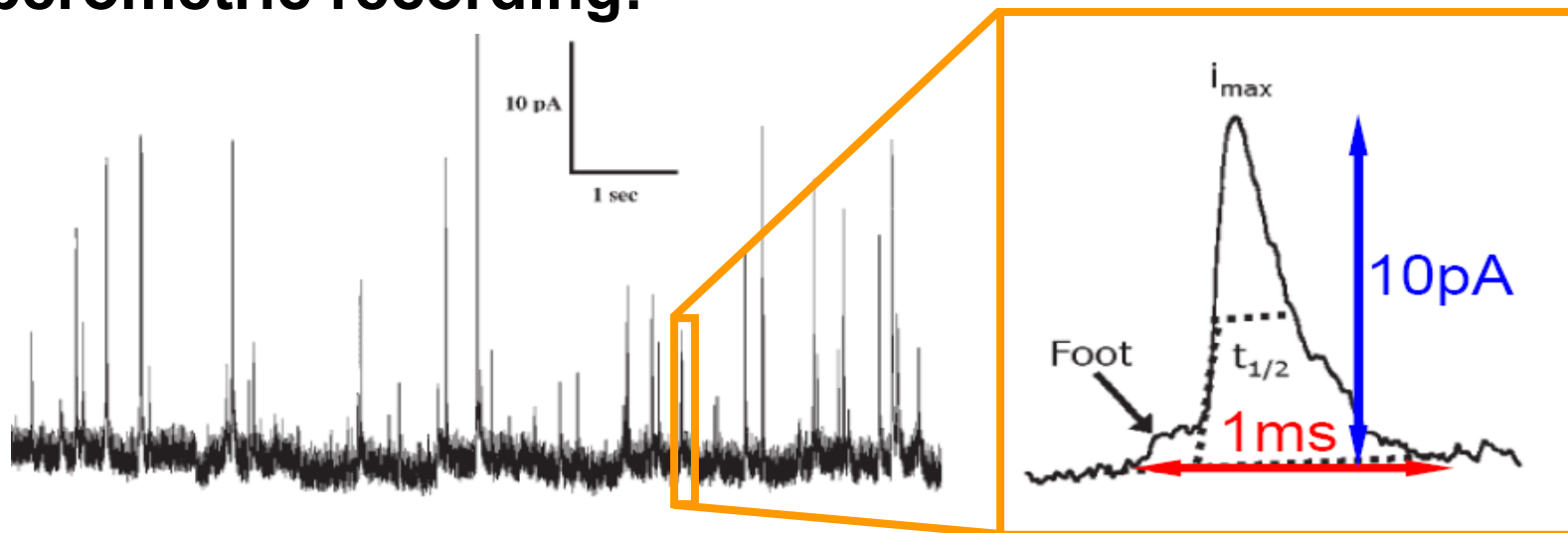
Combination of Techniques



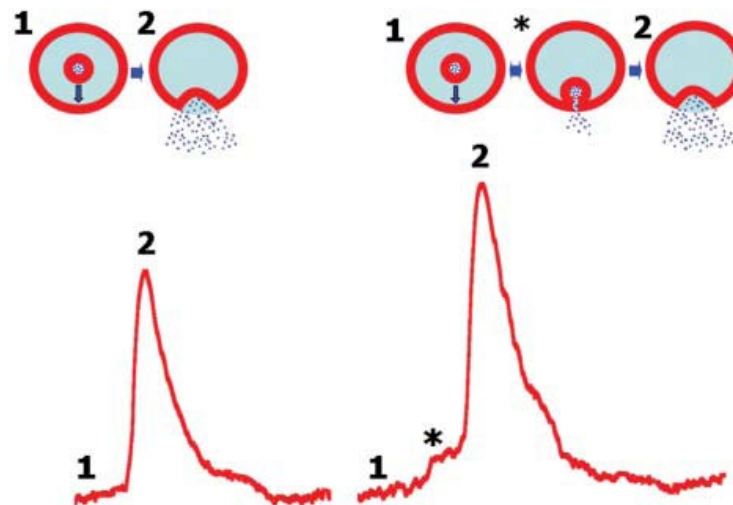
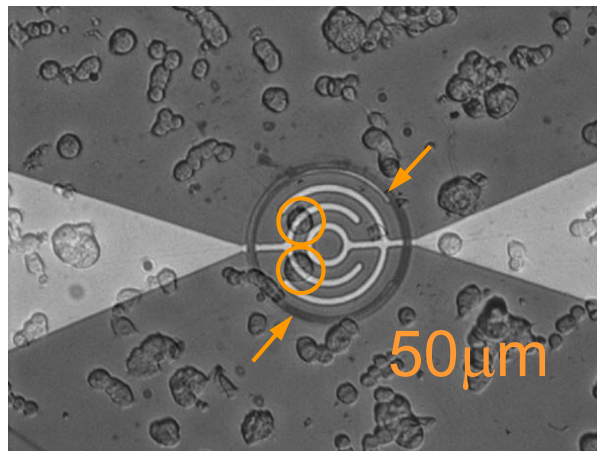


Detection of Single Exocytotic Events

Amperometric recording:



Single PC12 cells

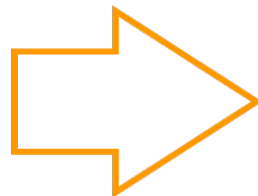


C. Spegel et al., *Electroanalysis* **19** 2007 & *Lab Chip* **8** 2008

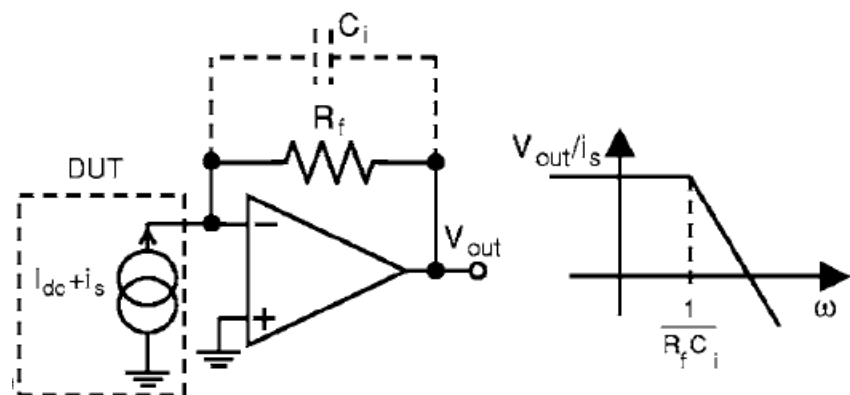
Specs: 1pA Resolution @ <1ms

At the **limits** achievable with a standard transimpedance amplifier:

- $R_f = 200\text{M}\Omega$
- 0.2pF stray capacitance



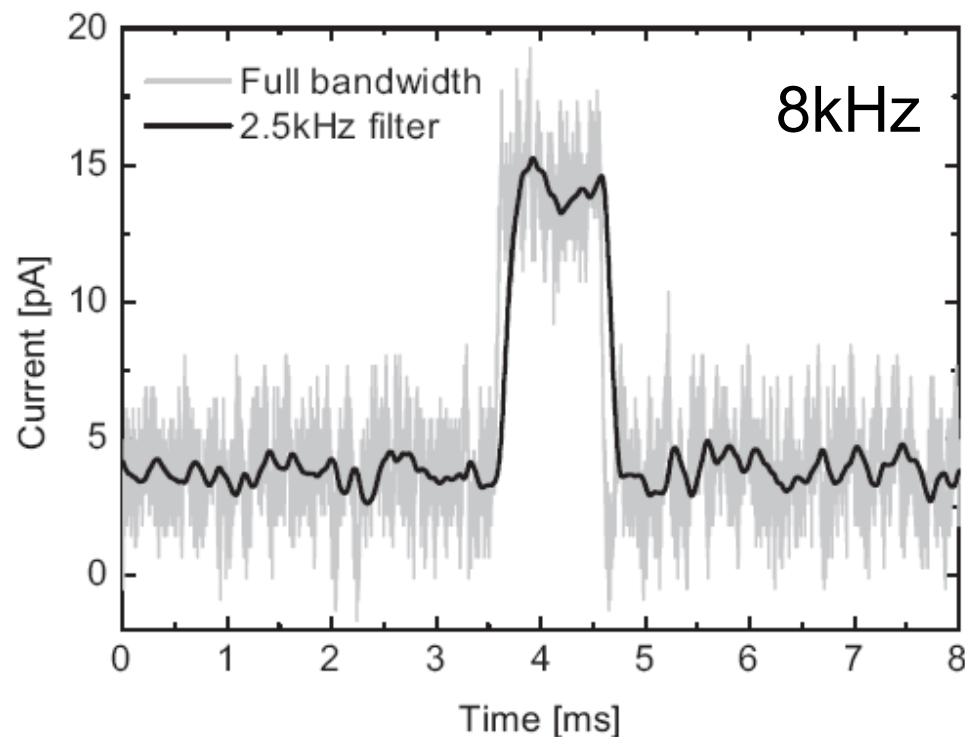
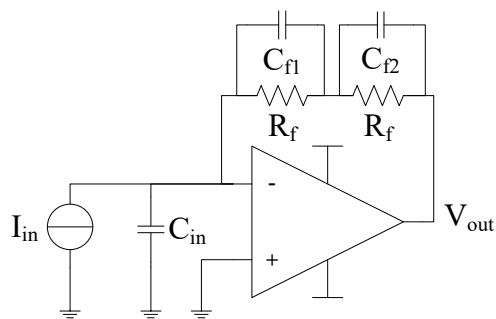
- $\text{BW} = 4\text{kHz}$
- $1.4\text{pA}_{\text{rms}}$



Solutions:

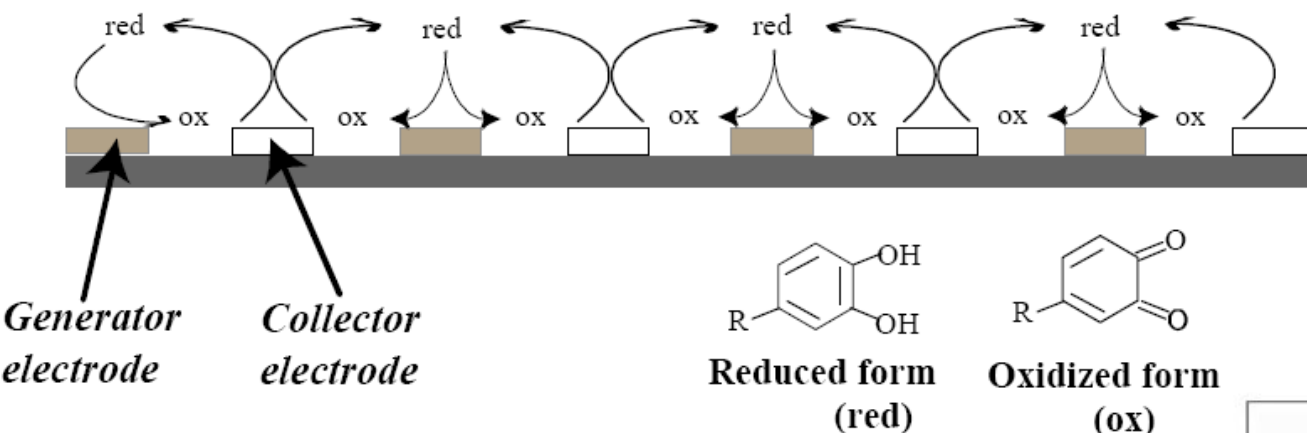
Integrators

- Switched
- $H(s)$



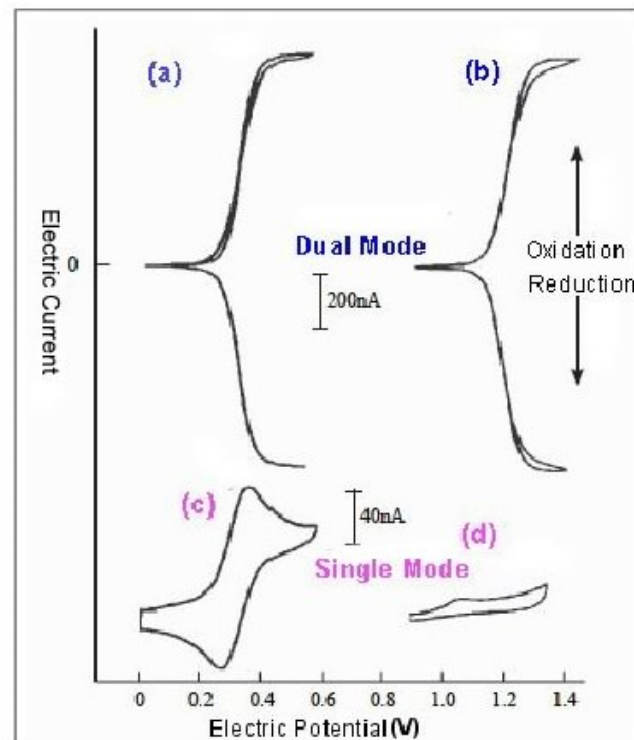
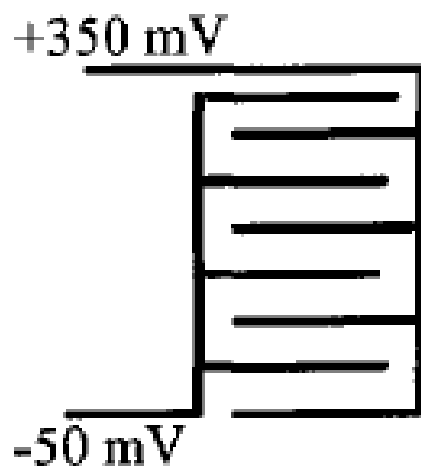


Intrinsic Amplification: Redox Cycling

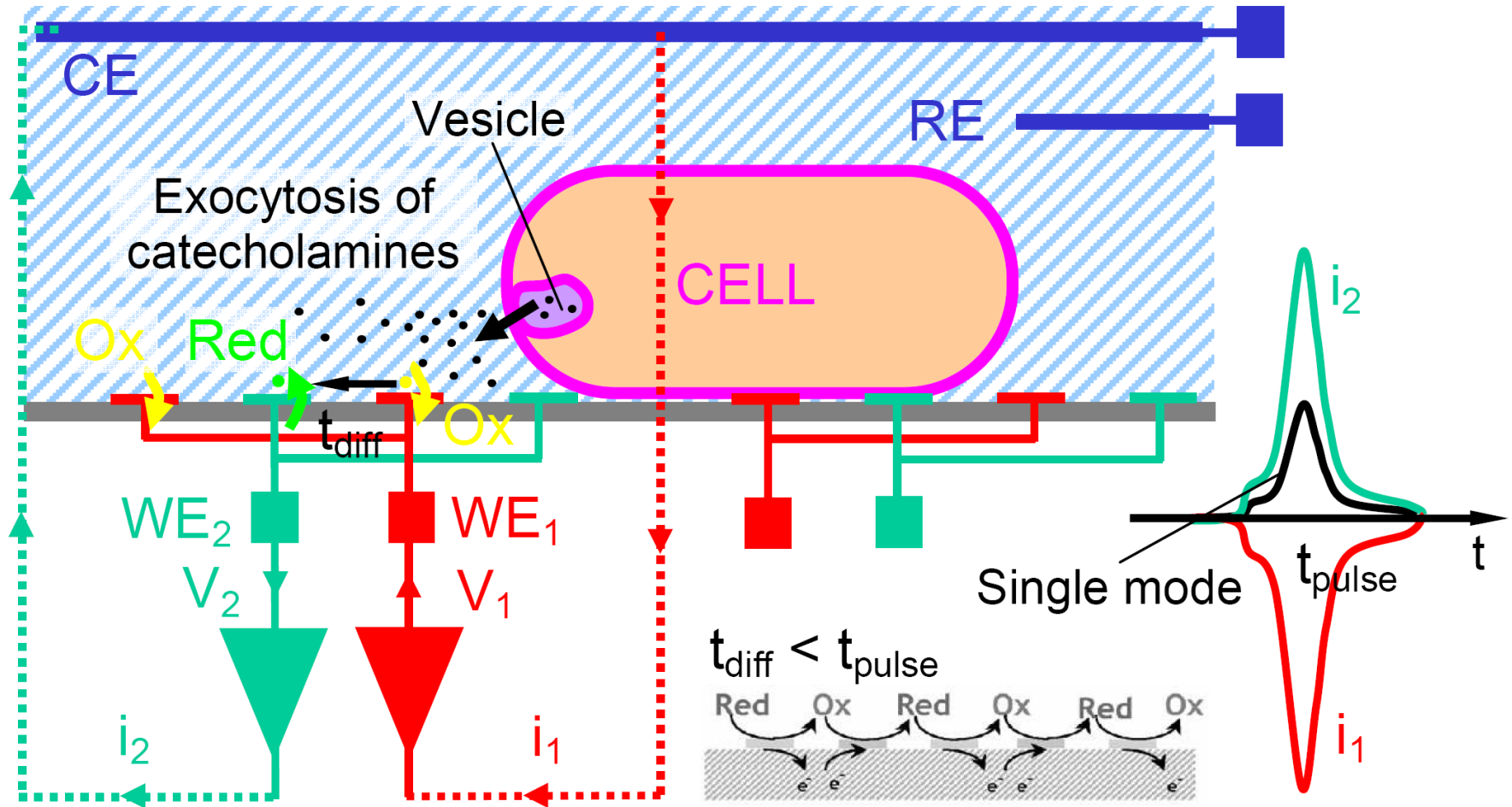


Must be reversible!

Planar
interdigitated
electrodes
spaced by μm



Paeschke *et al.* *Analytica Chimica Acta* **305** 1995



With *catecholamines* the achievable gain is ~ 10

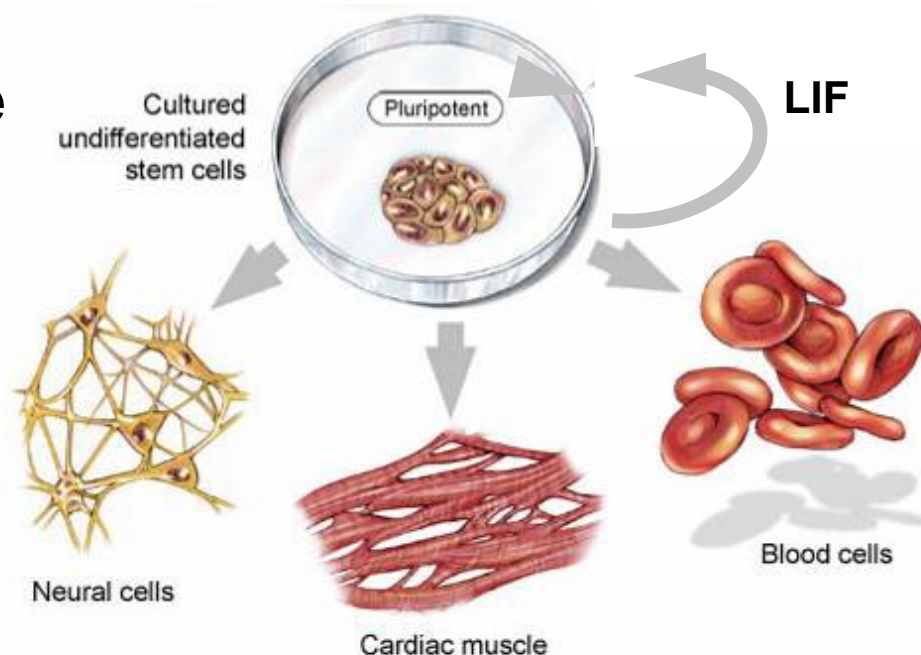


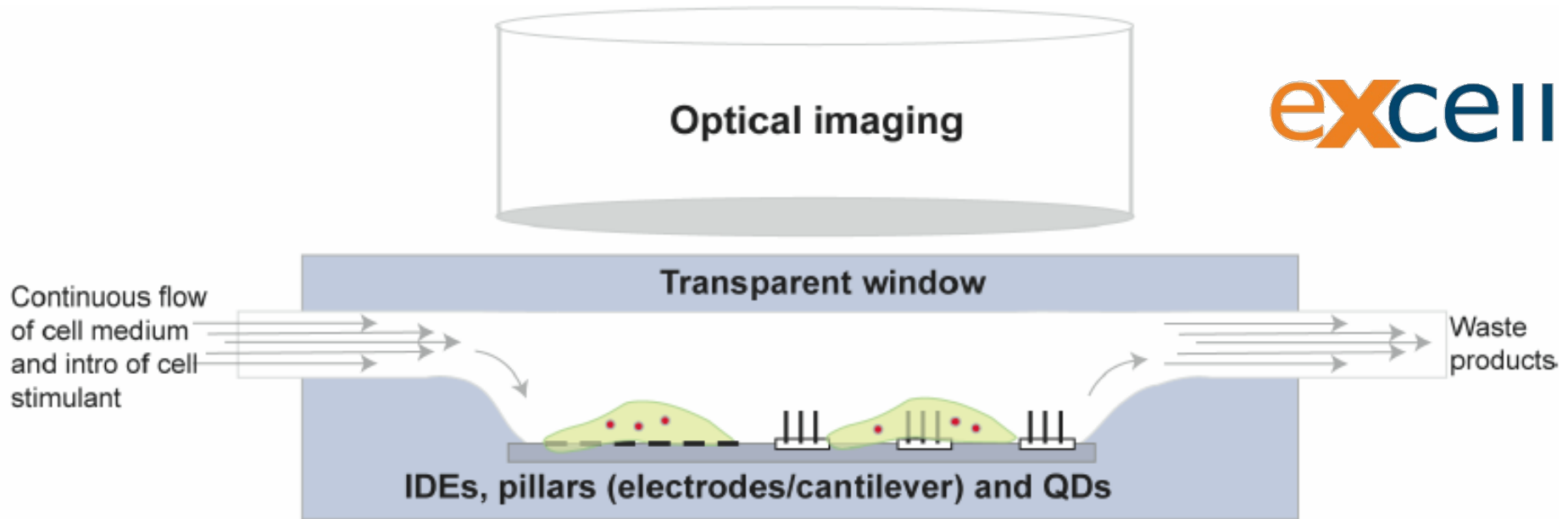
High-sensitivity tracking of neurotransmitter exocytosis:

- Investigate fundamental biological mechanisms (brain)
- Addressing neurodegenerative diseases
- Early detection of stem-cell differentiation

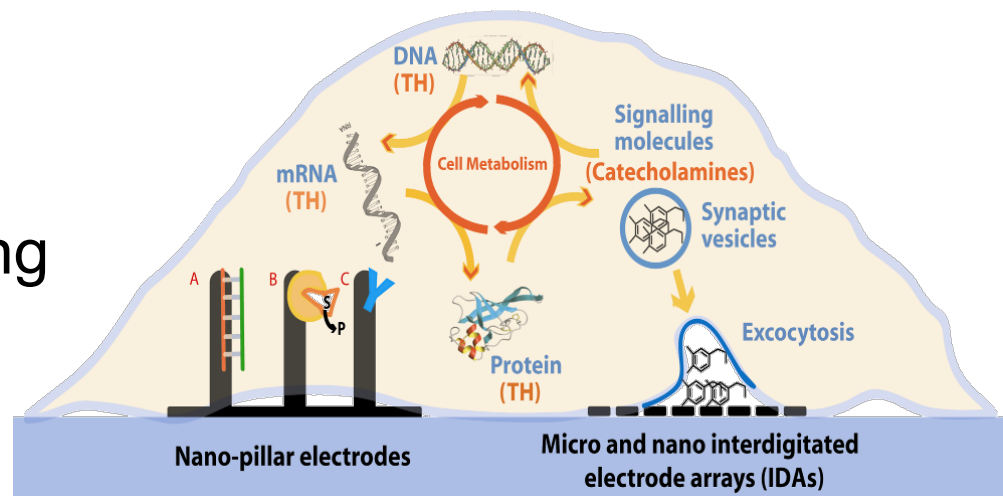
A major biological challenge

Understand and control the differentiation triggers and paths of stem cells





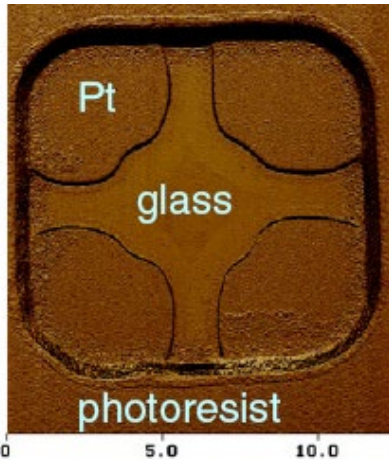
- Optical/electrical detection
- Microfluidics
- Intra/extra cellular recording
- Automatic
- Highly parallel



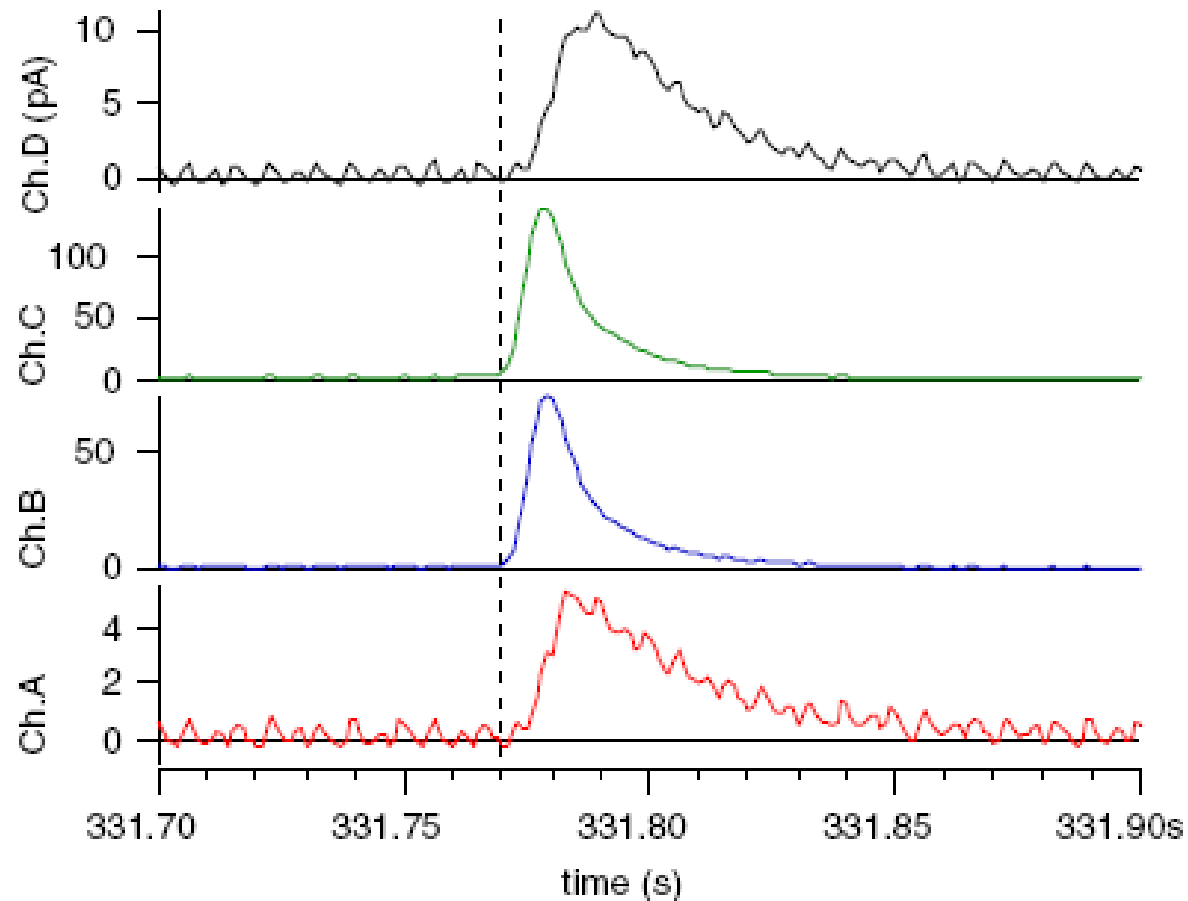
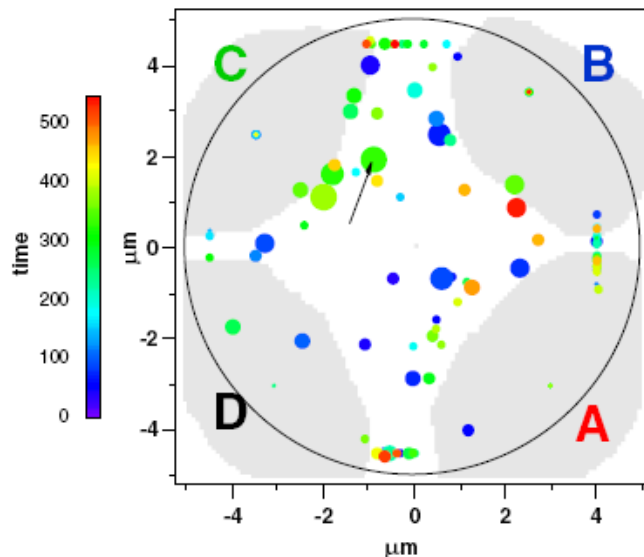


Sub-Cellular Resolution

High-resolution spatio-temporal tracking of single exocytotic events:



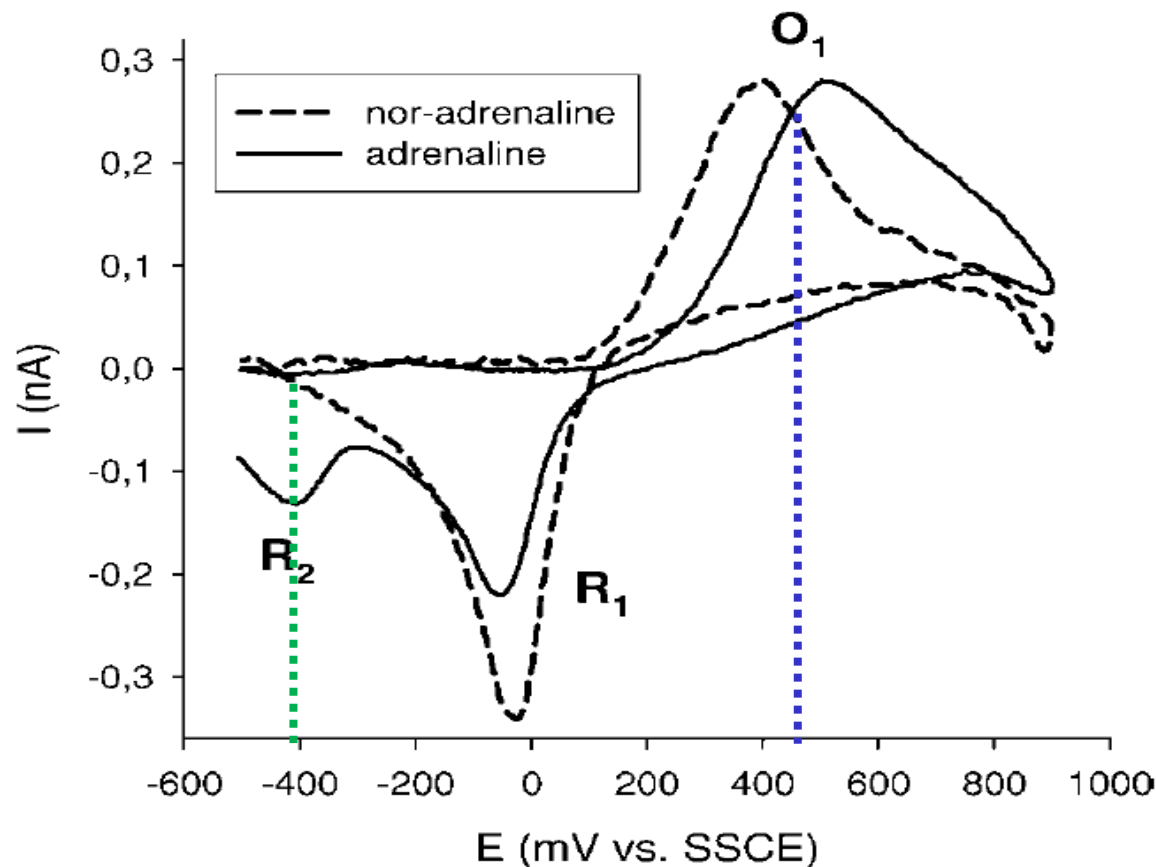
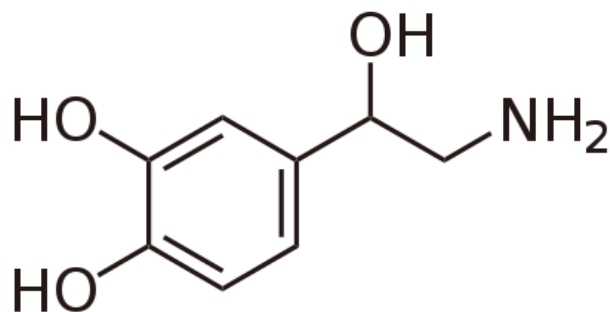
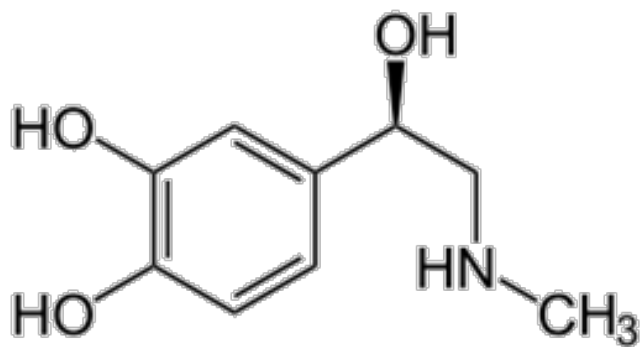
Single Cromaffin cell



A. F. Dias *et al.*, Nanotechnology **13** 2002

The Problem of Interferences

Similar neurotransmitters (similar molecules, similar CV) may have very different biological function → **selectivity issue:**



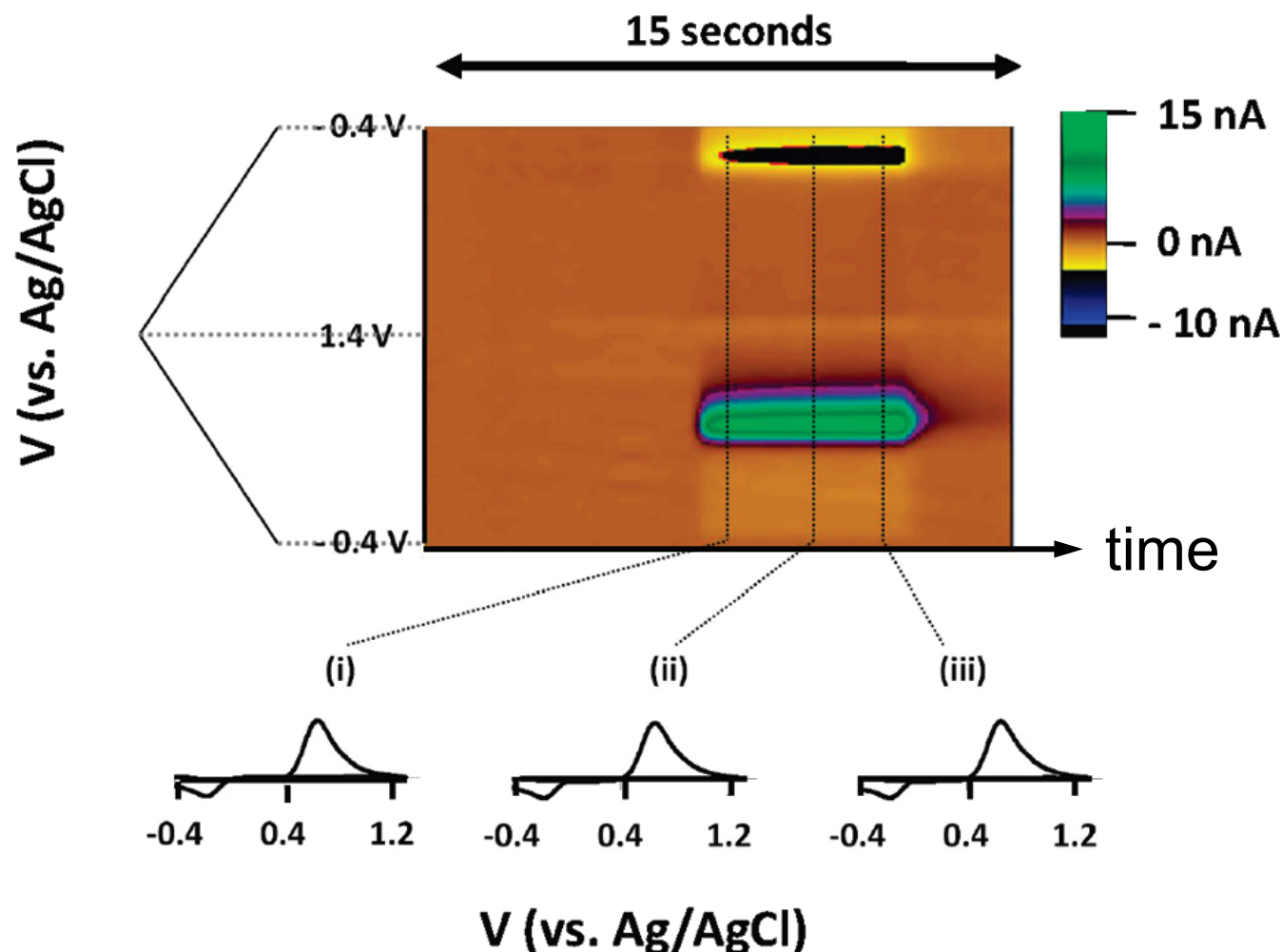


Solution: Fast Cyclic Voltammetry

In-vivo rat brain:

(1) Dopamine ($1\ \mu\text{M}$)

$400\text{V/s} \rightarrow 100\text{ms}$



M. Wightman, *Anal Chem*, **81**, 2009

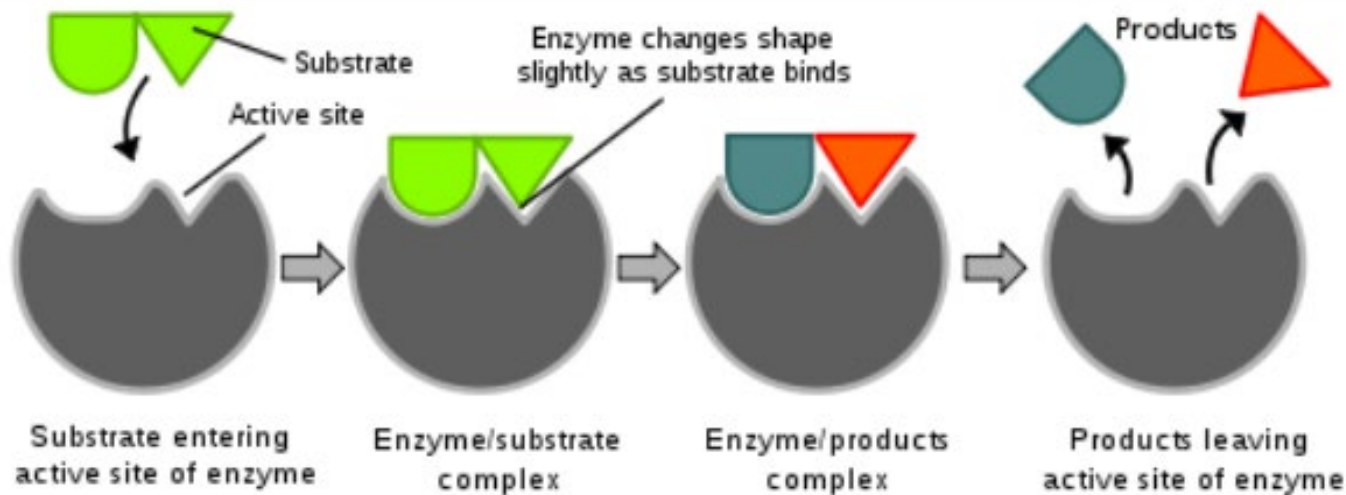


Extension of Amperometry

If the molecule is not electroactive?



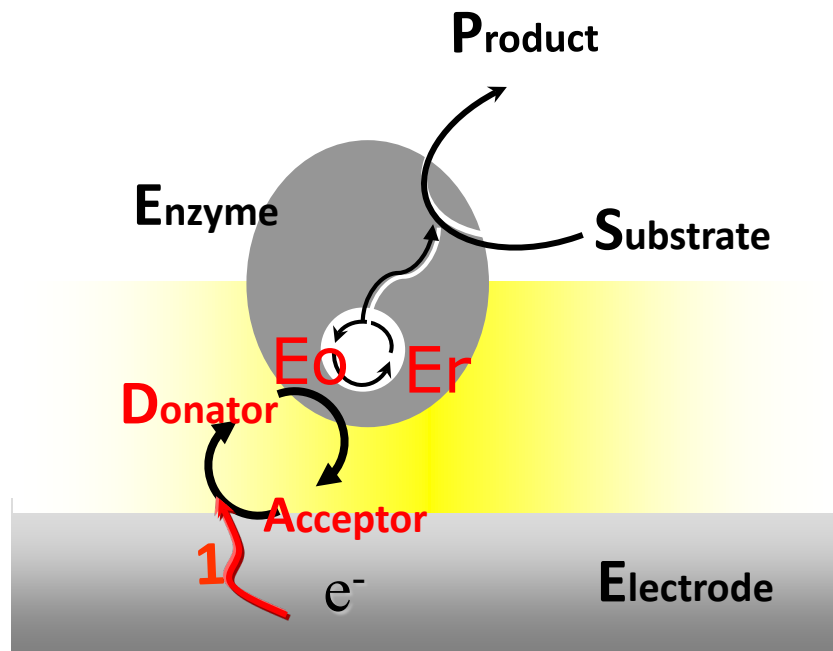
Use a specific mediator



Enzyme!



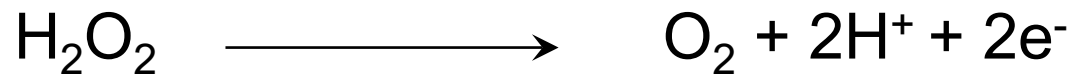
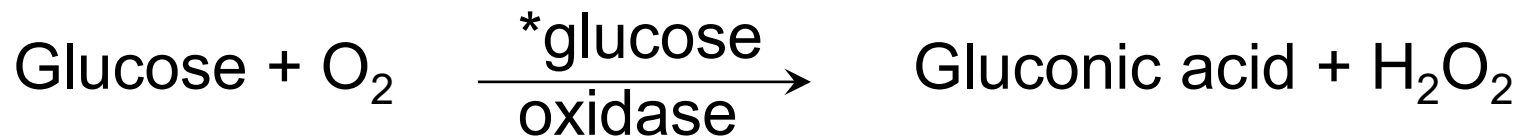
4. Enzymatic Amperometric Biosensor



If the catalysis of the target molecule involves an **electron transfer** it can be detected with amperometry

- The current is proportional to the concentration
- The process can be less efficient
- Enzymes are often immobilized on electrodes

Enzymatic Reactions for Glucose



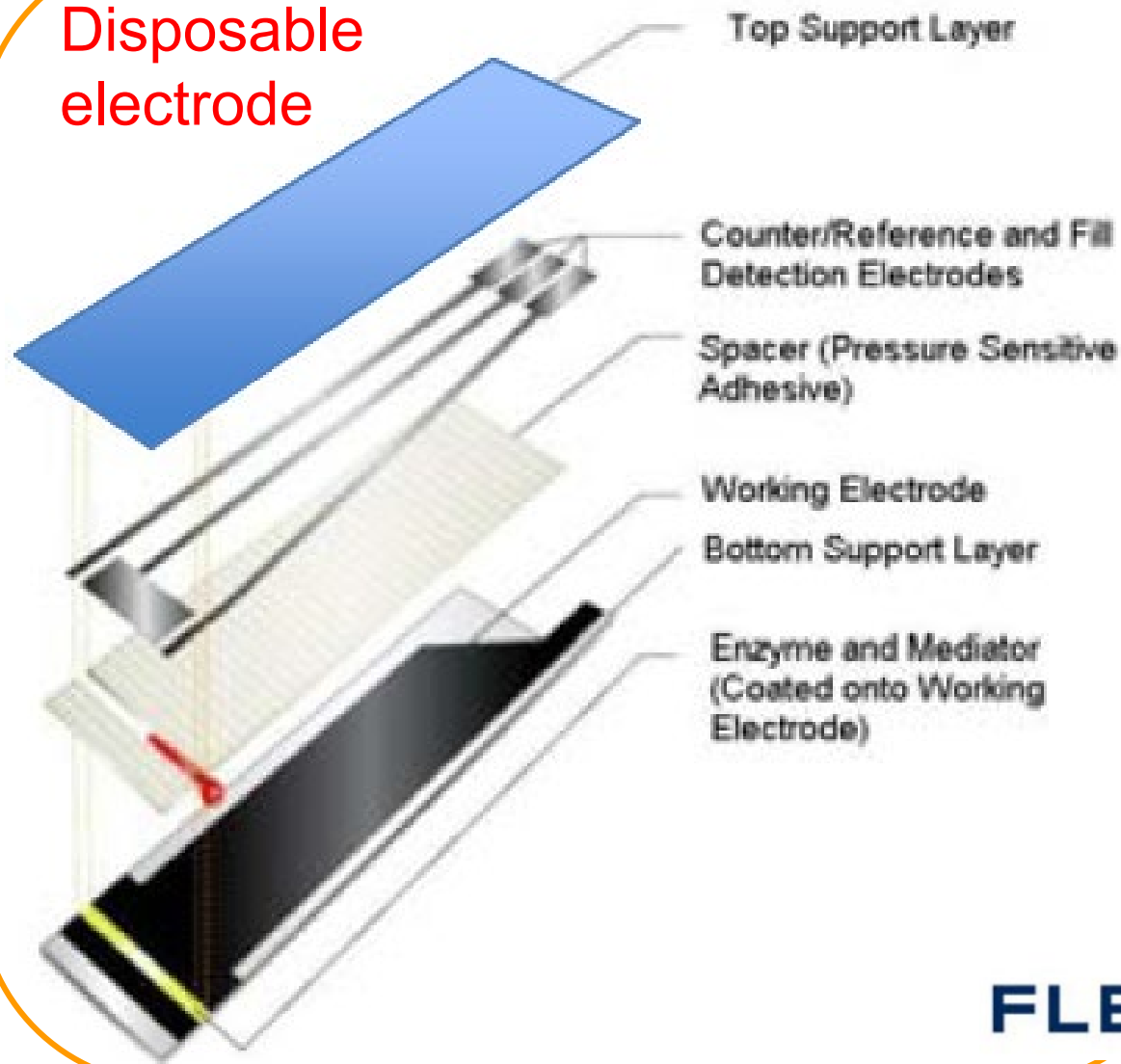
Total Reaction:



* Glucose oxidase (GOD) is commonly used since it is fairly stable & requires no cofactors or coenzymes

Commercial Biosensor System

Disposable
electrode

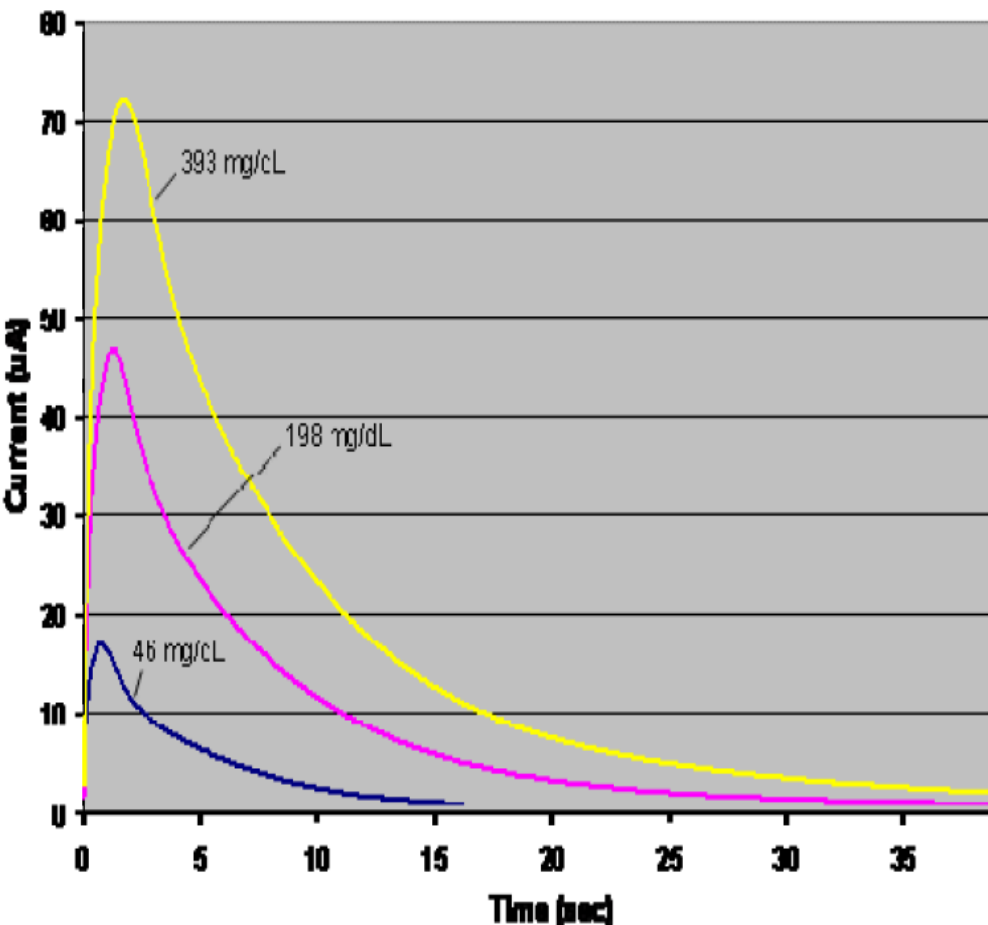


FLEXTRONICS X

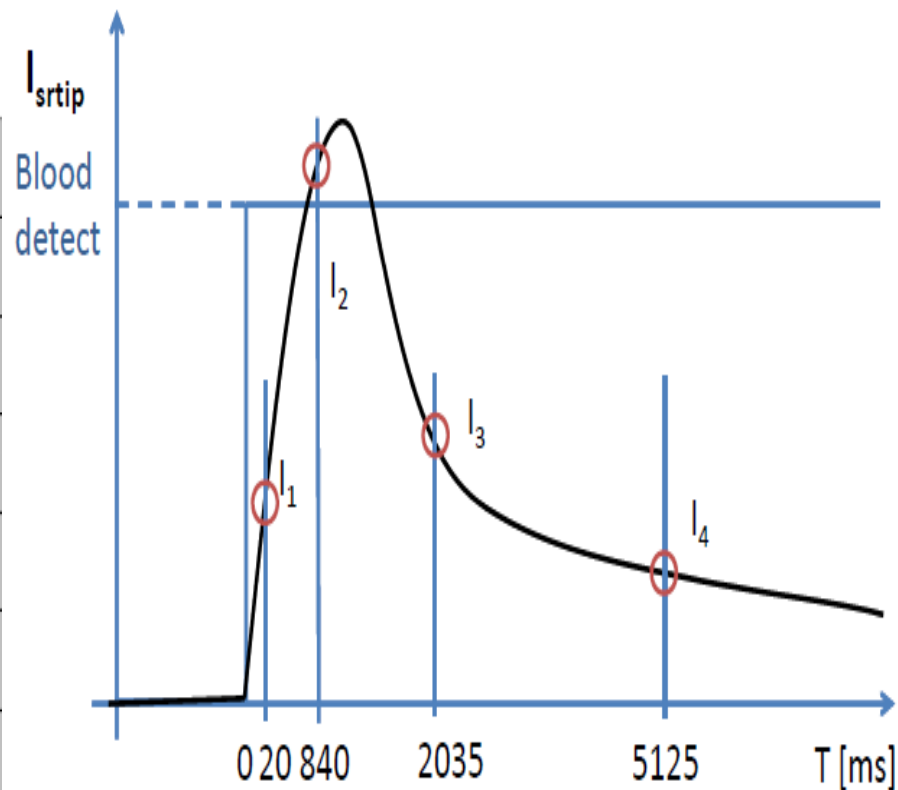


Real Curves

Electrochemical blood glucose test strips (EBGTS)



(<http://electrochem.cwru.edu/encycl/art-g01-glucose.htm>)



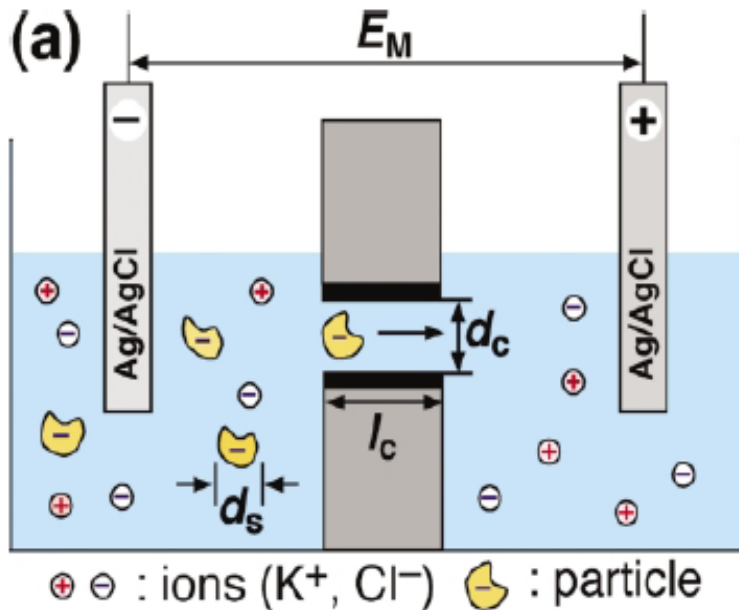
$$\text{Glucose}[\text{mg/dL}] = f(I_1, \dots, I_4, \text{Temp})$$

FLEXTRONICS



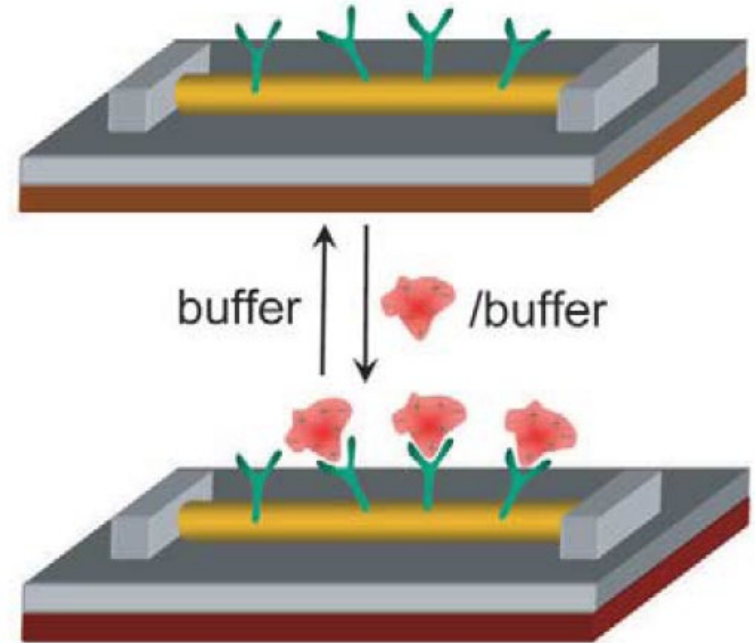
5. Nanoscaled Transducers

Nanopore sensor



- natural or artificial
- $\varnothing \sim 5\text{nm}$

Nanowire sensor



- diameter $\sim 10\text{nm}$, $\sim \mu\text{m}$ long
- electrostatic effect and/or conformational changes

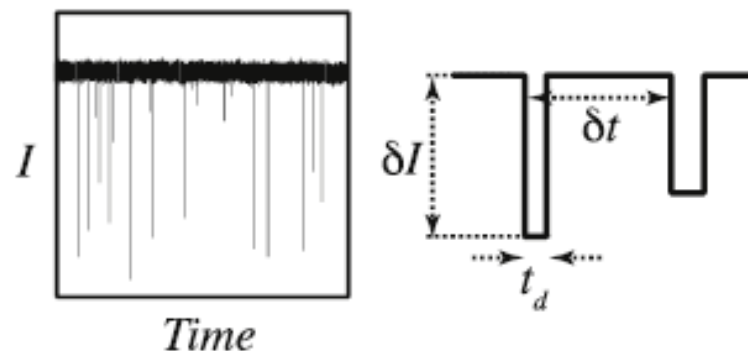
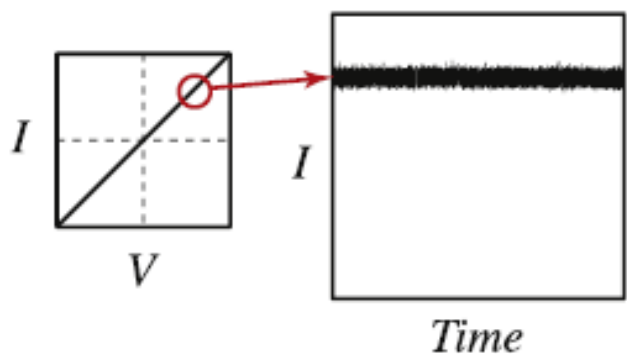
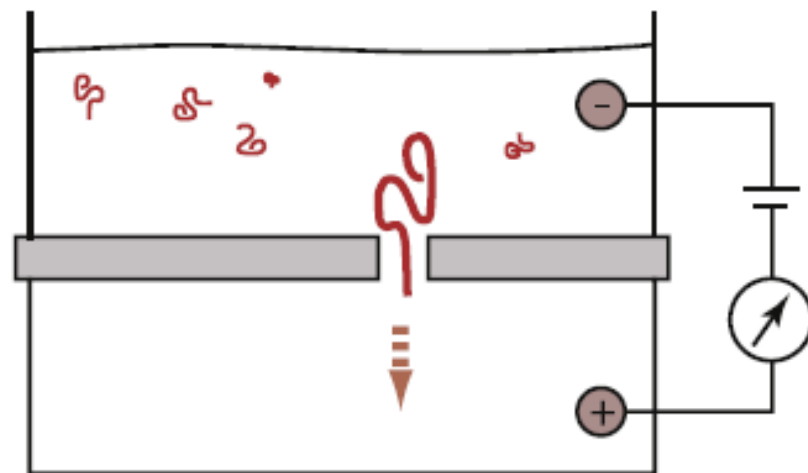
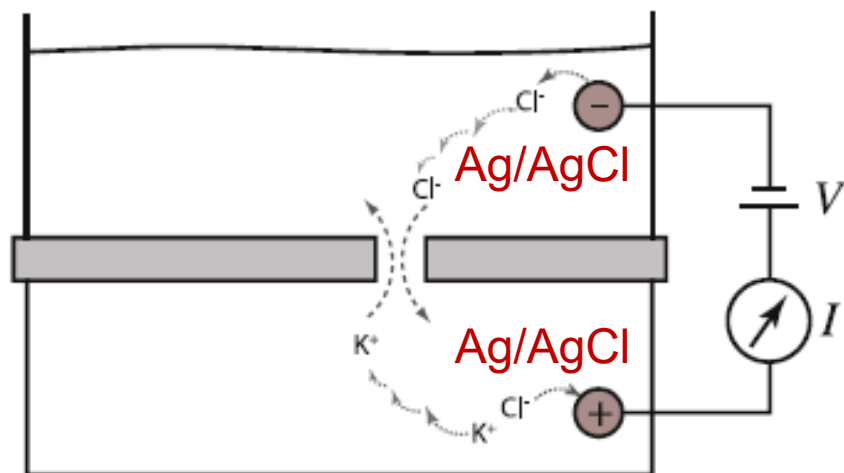


change in the conductance



Recording Current in Time Domain

Detection of the obstruction of the pore when a molecule passes:

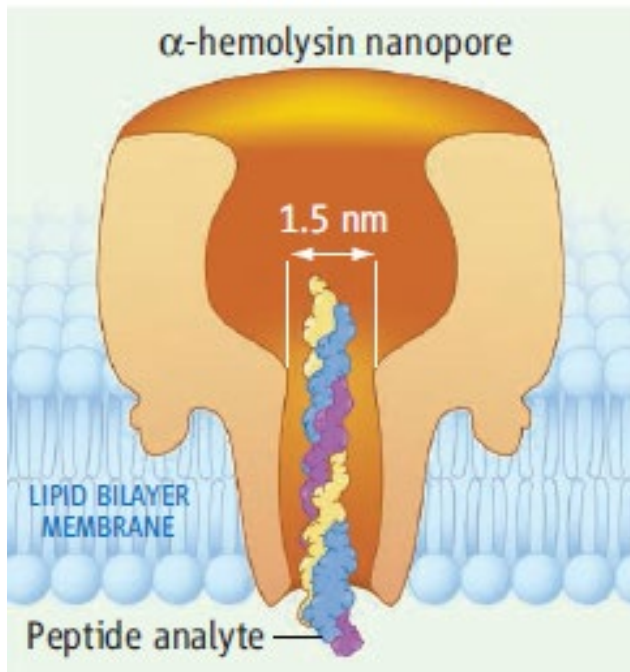




Explosion of Nanopore Technology

Mimicking natural detection scaling down the pore at nanometers:

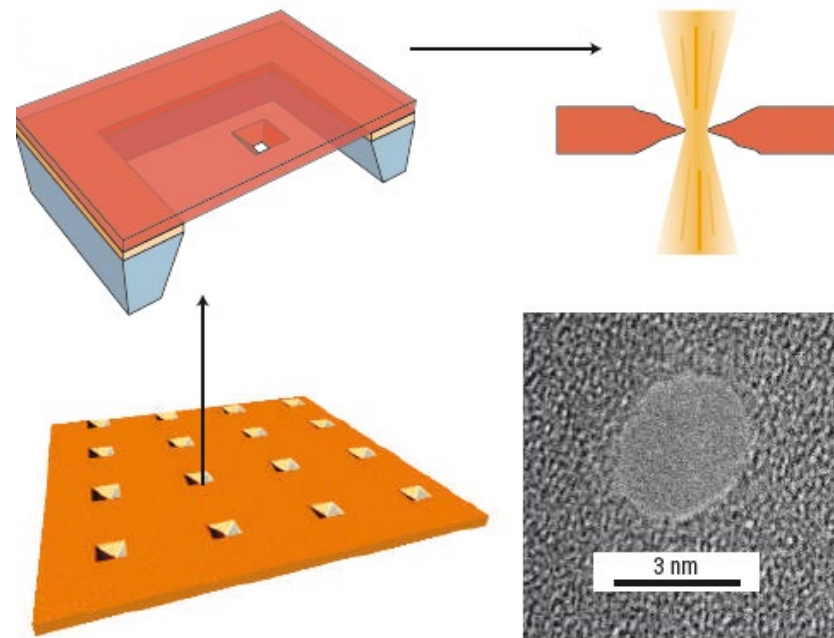
Natural



Immobilization and stability issues.

Martin, *Science* **317** 2007

Solid-State

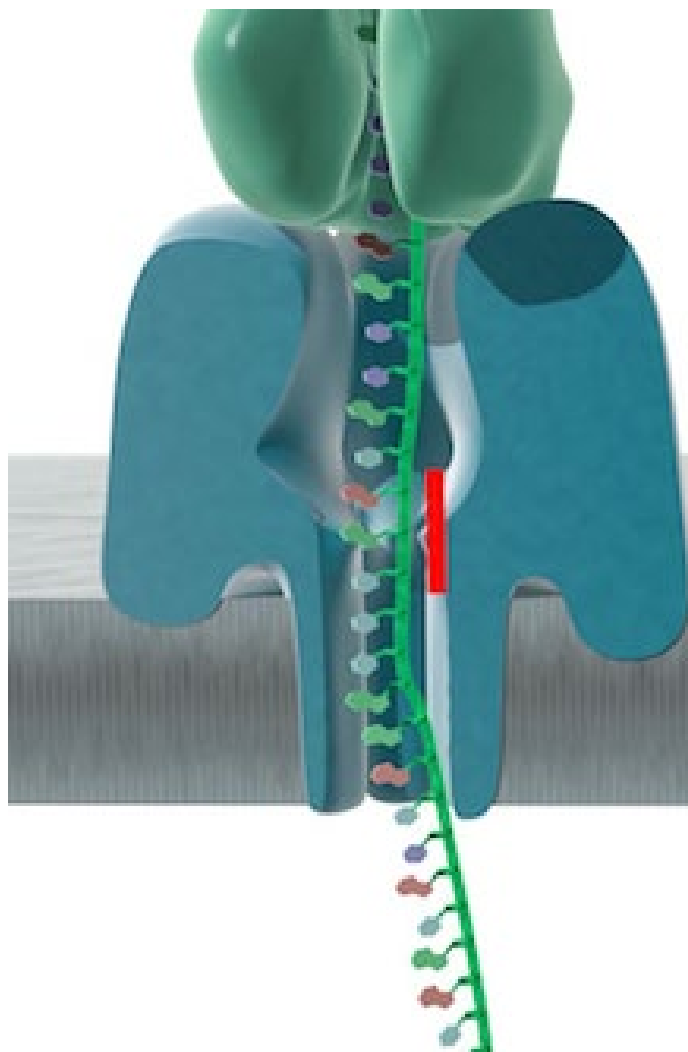


Fabrication with ion (FIB) or electron beams (TEM).

Dekker, *Nature Nanotech.* **2** 2007



DNA Sequencing



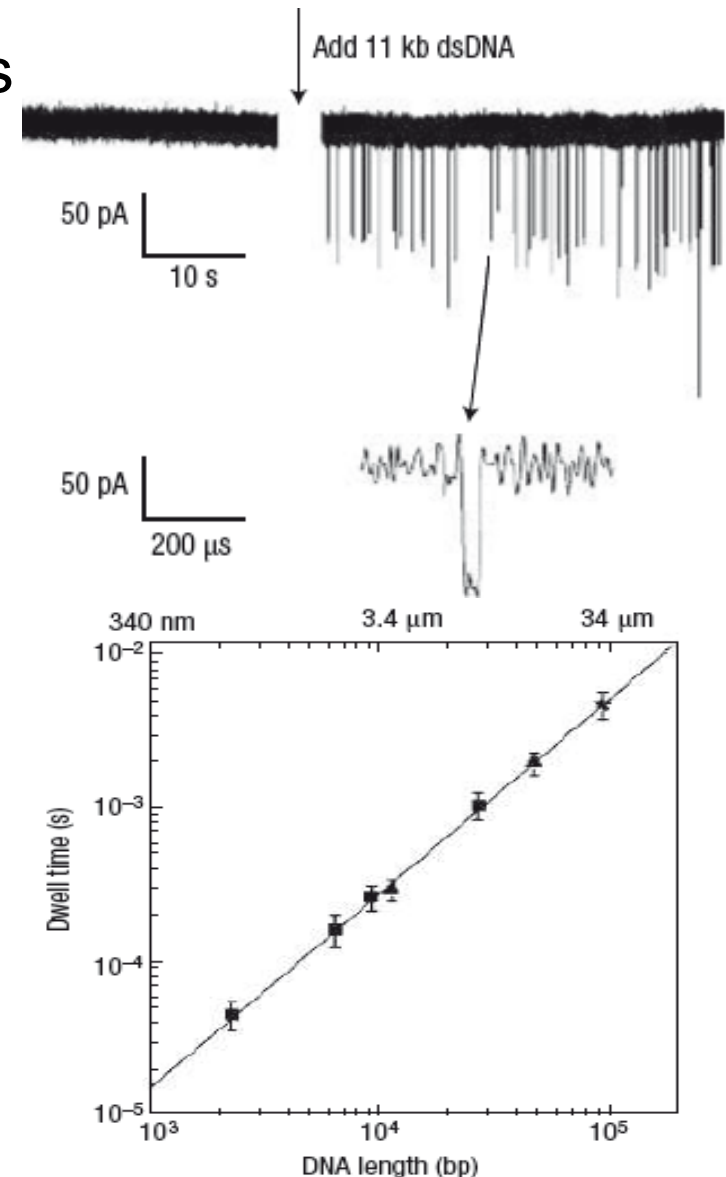
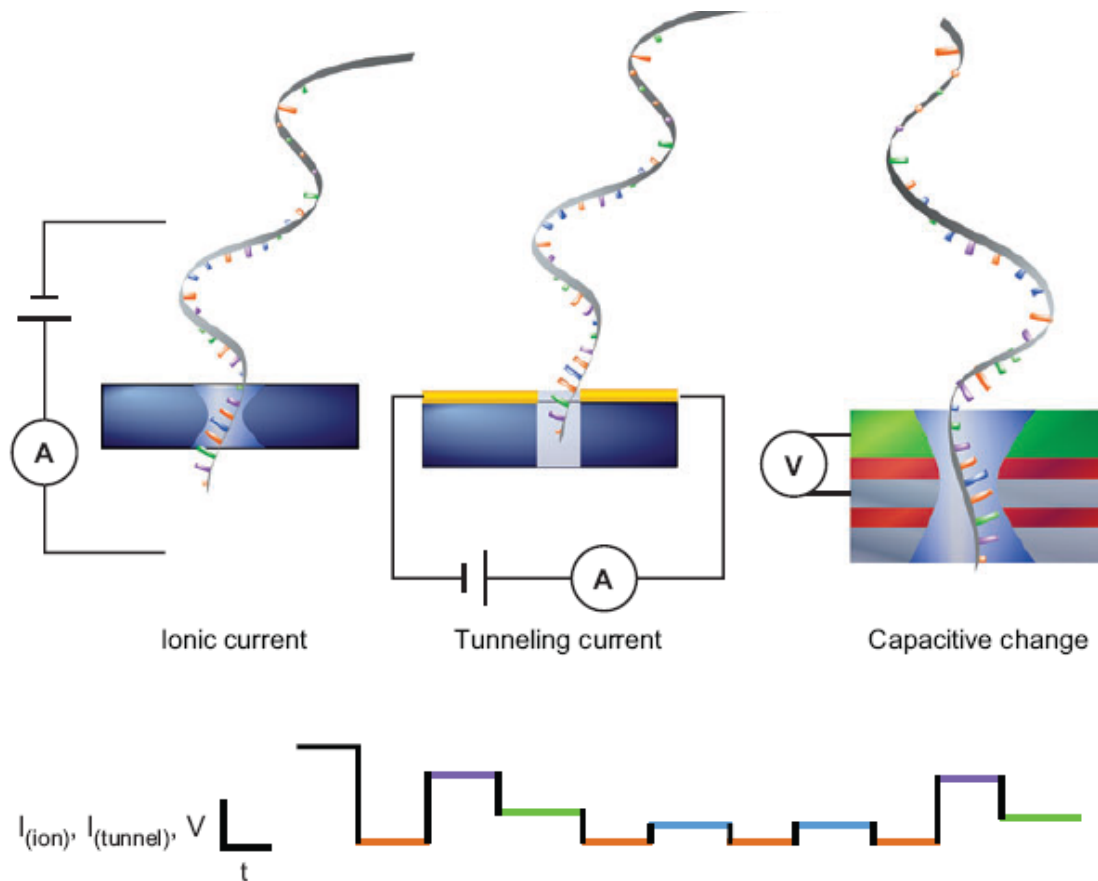
- Simple
- Personal / Point-of-Care
- Low-cost
- Automatic
- Fast





Towards Nanopore DNA Sequencing

Use the pore to detect single DNA bases
Thickness issues

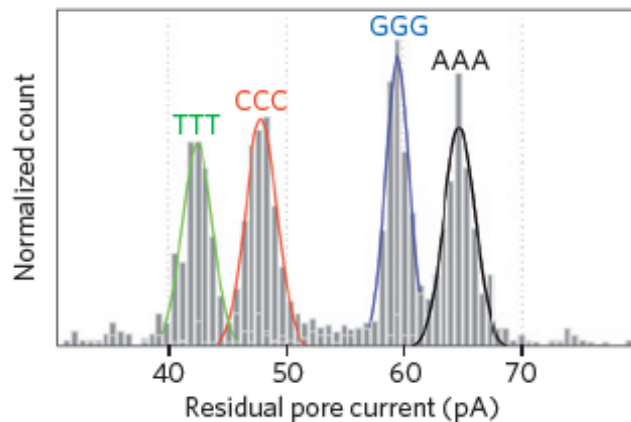
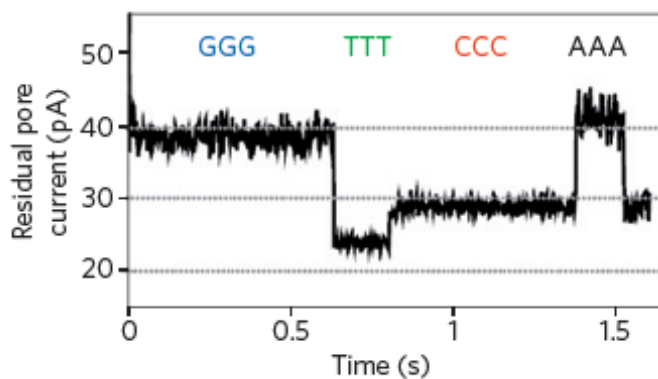
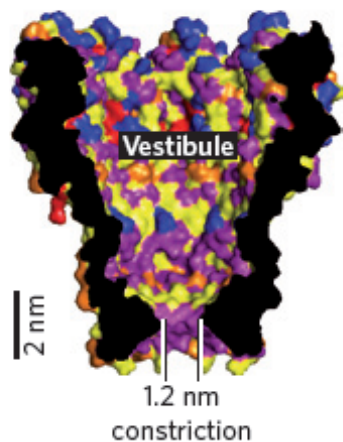
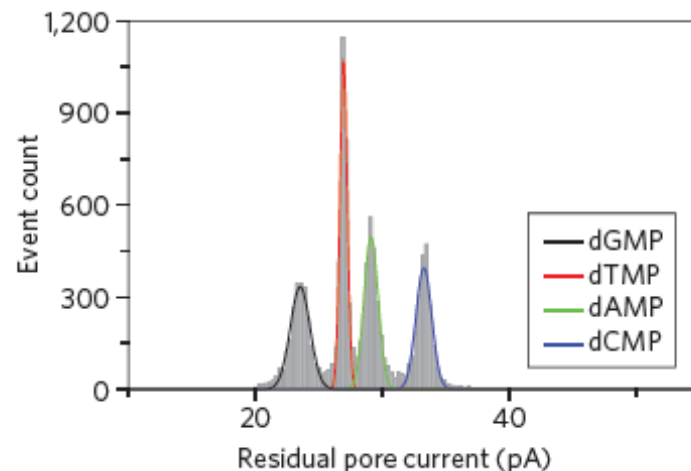
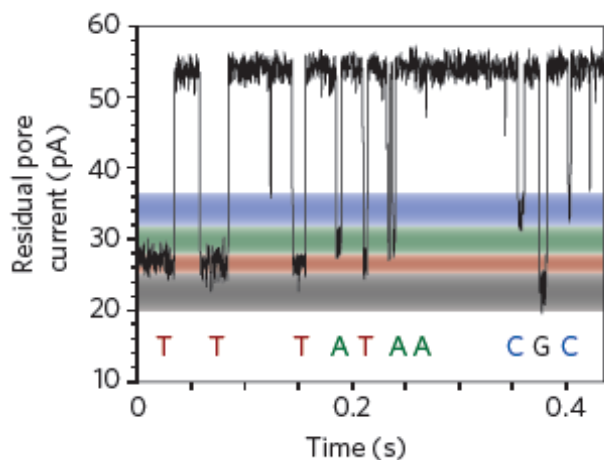
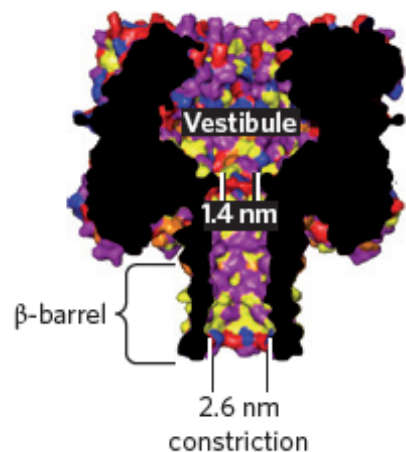


D. Branton *et al.*, *Nature Biotech* **26** 2008



Encouraging Results

Capability to discriminate different monobase sequences:

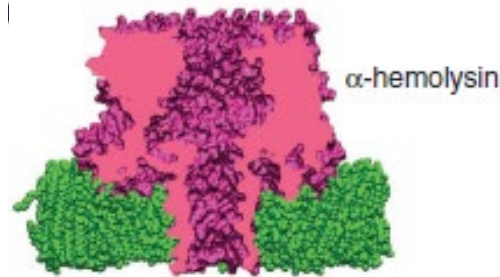




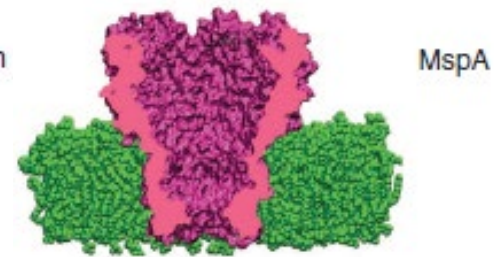
Specifications for the Current Detector

Pore blockade produces **current modulation** of:

- **100pA** for α -hemolysin
- **300pA** for MspA
- **1-4nA** for solid-state pores



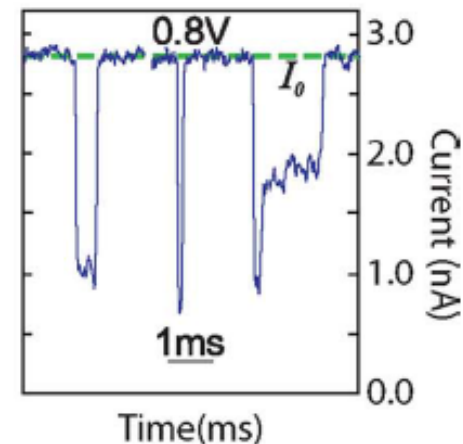
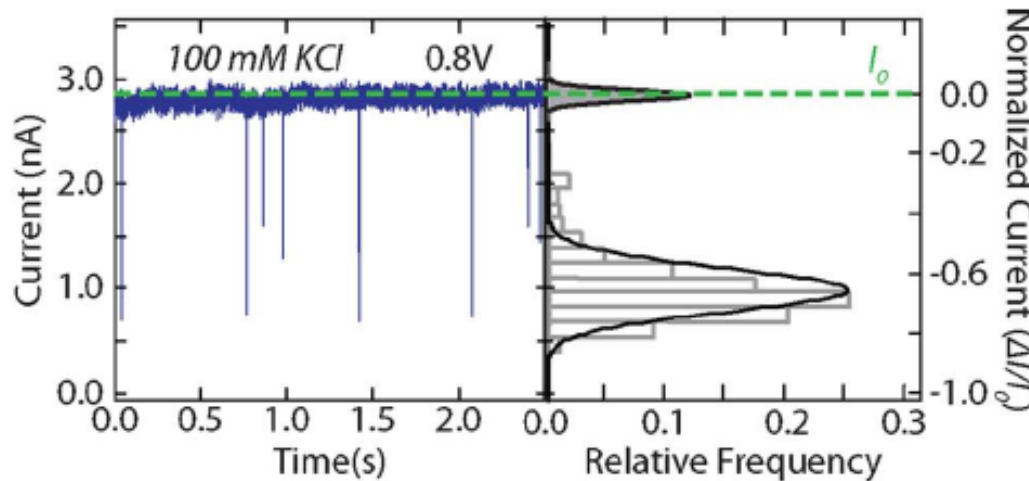
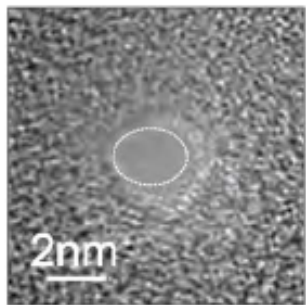
α -hemolysin



MspA

depends on: salt concentration, pore diameter, molecule type

Tailor size:



Timp, 2010

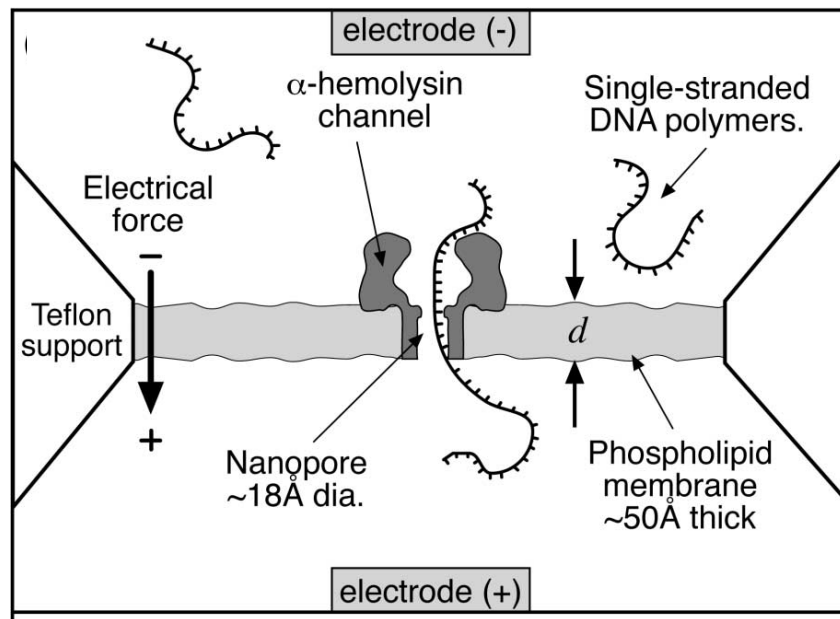


Specifications: Translocation Speed

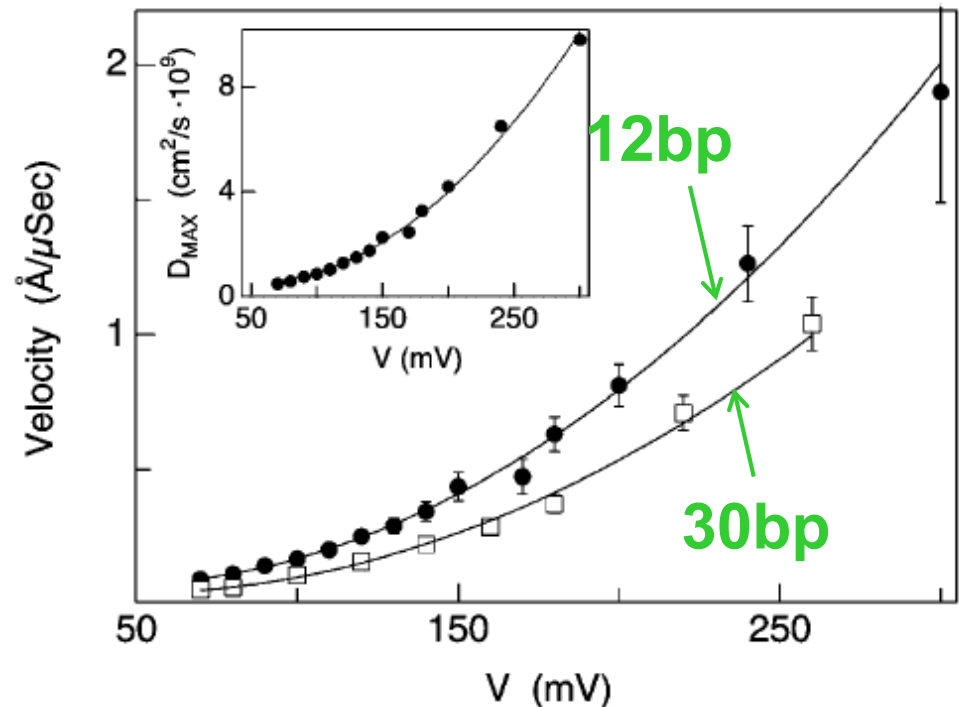
DNA ($-q$) translocation is controlled by **electrophoresis**

Speed depends on the voltage V and on the polymer length (for short chains) and on the kind of pore, typically:

- **1 bp/1 μ s** for α -hemolysin
- **25 bp/1 μ s** for solid-state pores



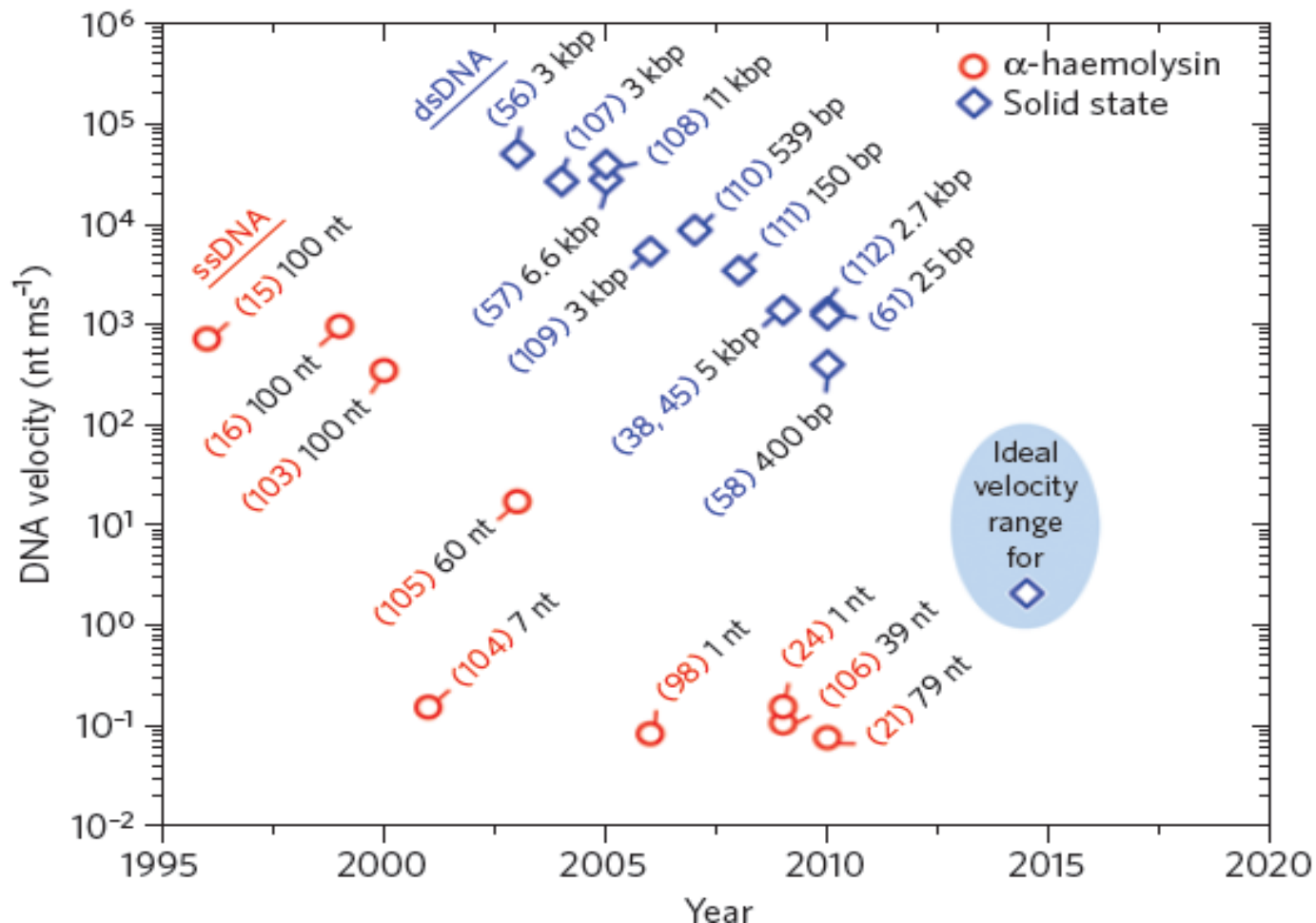
A. Meller *et al.*, *PRL* **86** 2001





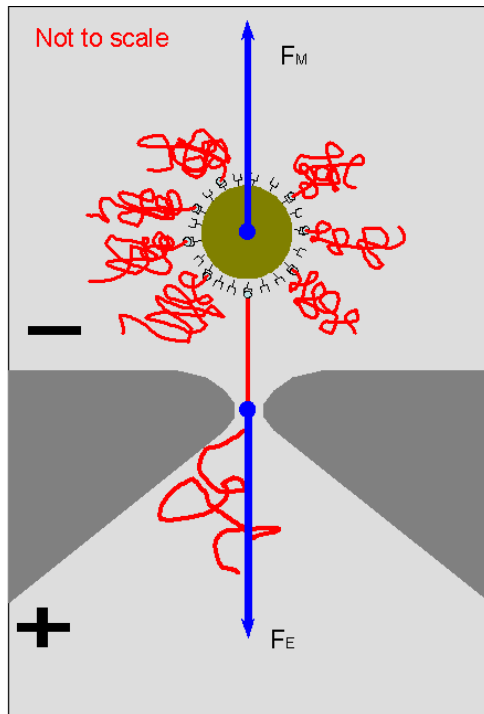
Solution: Take More Time

To have ~ 10 pA resolution, the BW is reduced to the kHz range
Translocation speed **needs to be reduced**



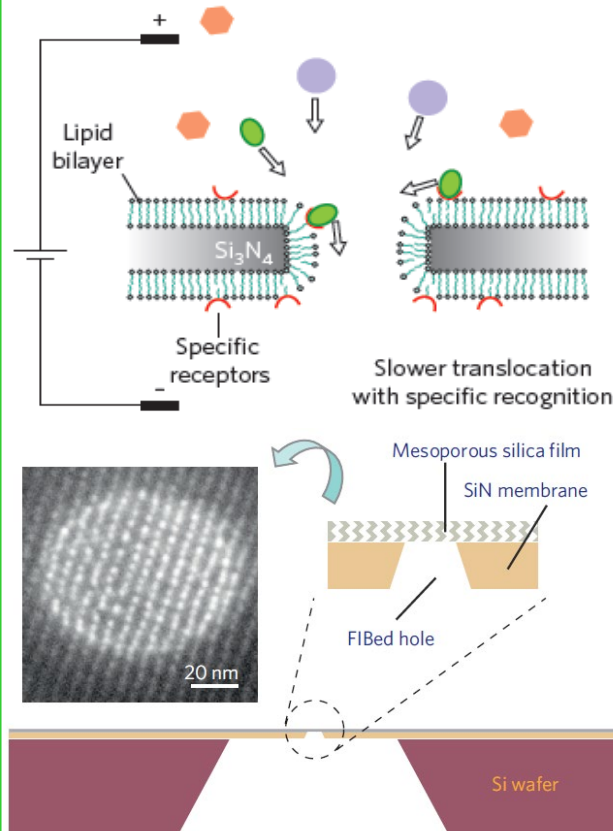
Approaches To Slow Down DNA

Magnetic tweezers



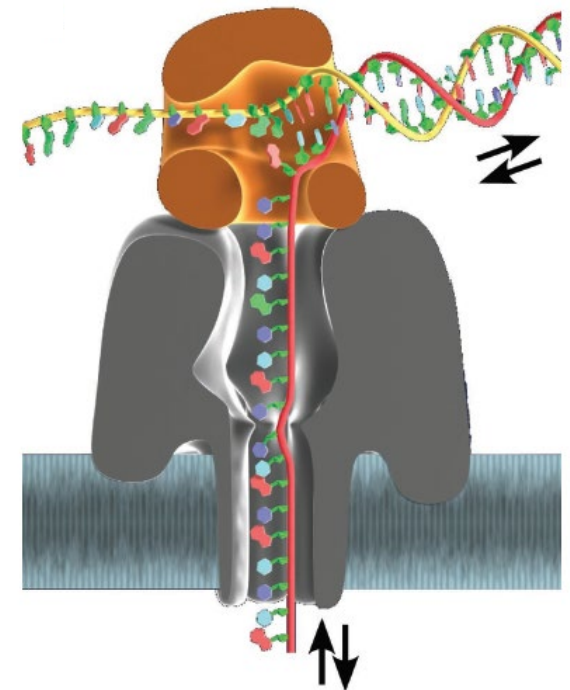
H. Peng & X. Ling,
Nanotechnology **20**
2009

Functionalized pore



Z. Chen et al., *Nature Materials* **9** 2010

Enzyme gateway



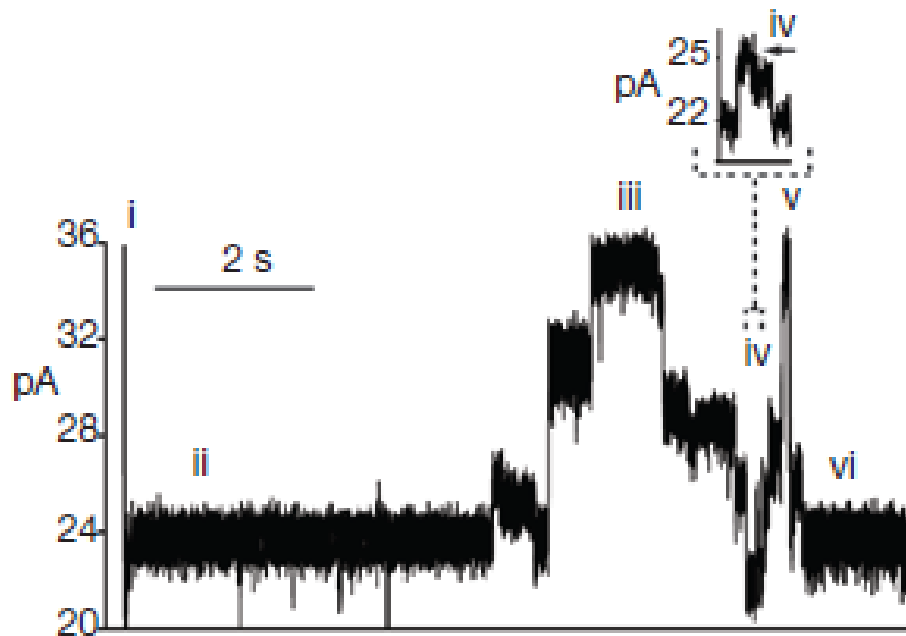
G. Cherf et al., *Nature Biotechnology* **30** 2012



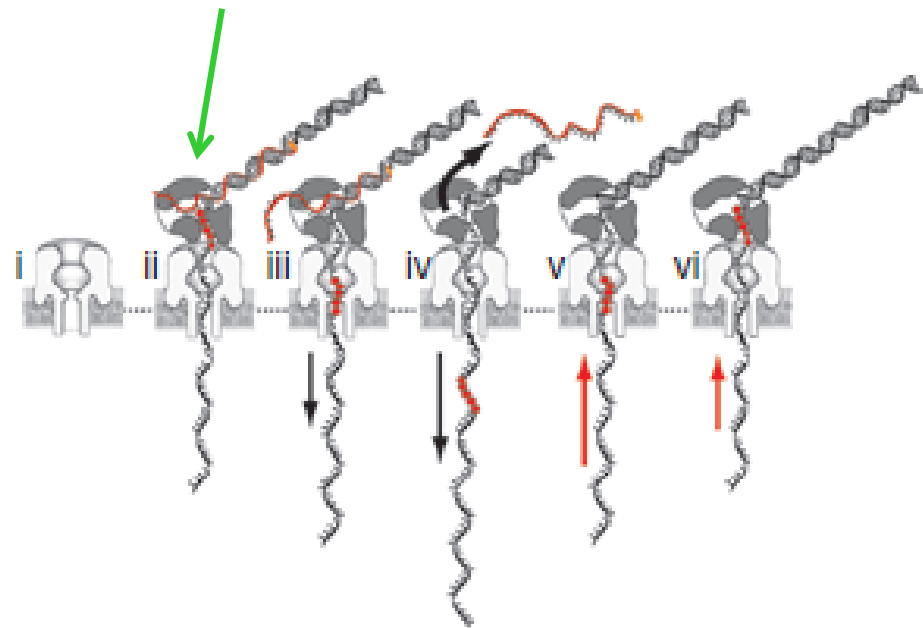
Enzymatic DNA Control

Automated forward and reverse ratcheting of DNA in a nanopore at 5-Å precision

Gerald M Cherf, Kate R Lieberman, Hytham Rashid, Christopher E Lam, Kevin Karplus & Mark Akeson



Enzyme: phi29 polimerase

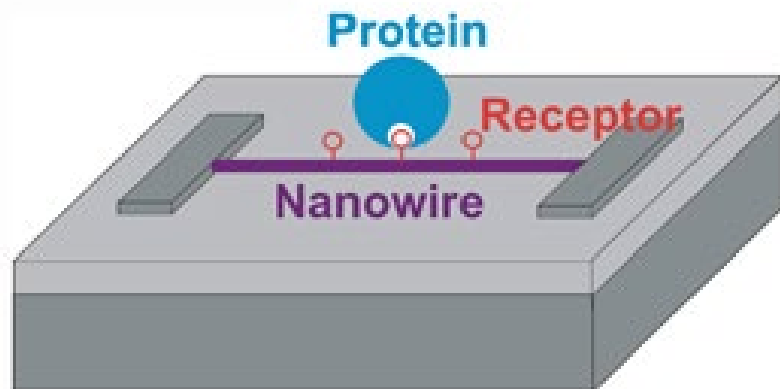


Impressive, but in the opposite to **high-throughput analysis**



Nanowire Biosensors

a

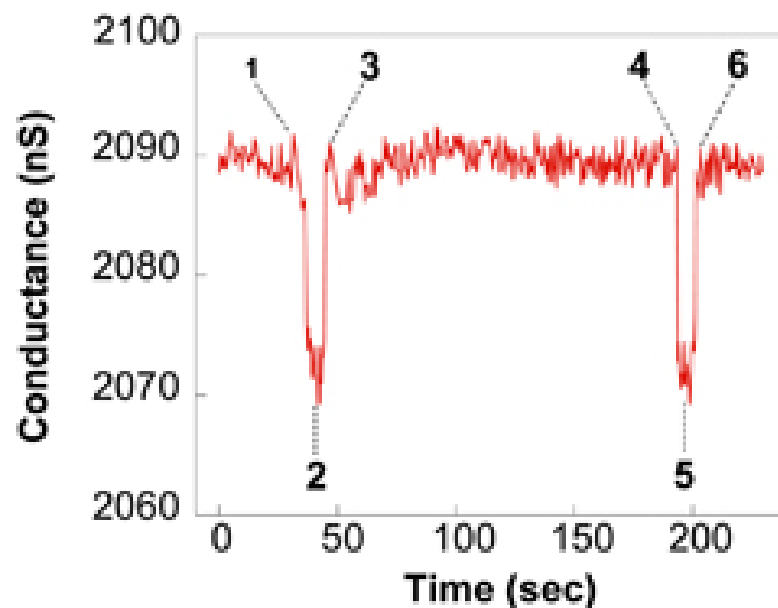


Lock-in technique: 30mV modulation of drain-source voltage ($\Delta I \approx 100\text{pA}$)

fM detection of PSA
(prostate specific antigen, cancer marker)

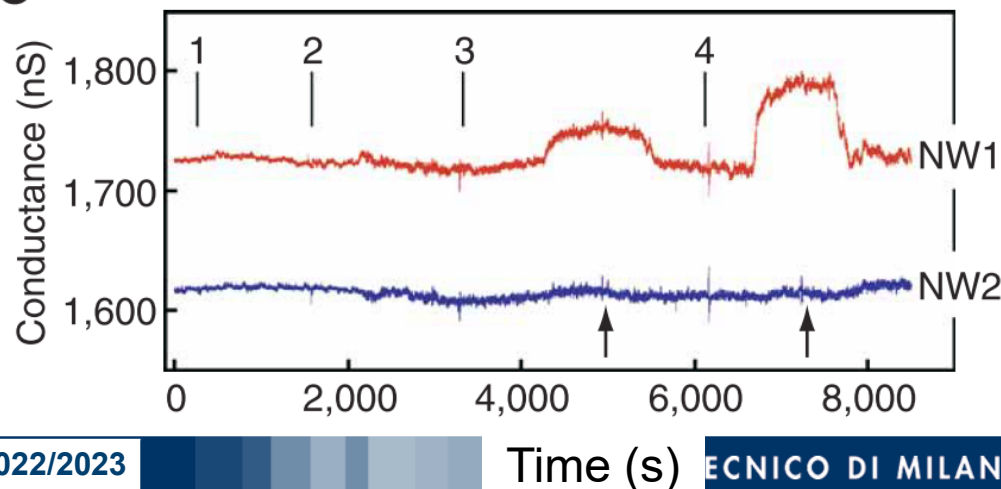
C. M. Lieber et al, *Nature Biotech.*, 2005, 23, 1294

b



Detection of single viruses

c



C. M. Lieber et al., Electrical Detection of Single Viruses, *PNAS*, 2004, 101, 14017.